Modeling Serialized Content Consumption: Adversarial IRL for Dynamic Discrete Choice

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Abstract

Serialized content—such as episodic fiction, TV series, or educational courses—presents consumers with a dynamic choice: pay for immediate access, wait for free access, or quit. These decisions are shaped not only by platform policies but also by the content itself. Understanding how these factors jointly influence consumption is critical for platforms aiming to optimize engagement and monetization. Content, however, is high-dimensional—capturing multiple layers of meaning and narrative structure—posing challenges for traditional estimation methods. We address this challenge by extending adversarial inverse reinforcement learning (AIRL)—a GAN-based estimator that recovers reward and policy functions from observed behavior. Methodologically, we (i) incorporate unobserved consumer heterogeneity, (ii) prove identifiability of action-dependent utilities, and (iii) develop a scalable approach to representing unstructured content that is suitable for adversarial reward learning.

We apply the model to 15 months of reading data from a serialized fiction platform (24,000 users, 6,000 chapters, 151 books). Estimates reveal two dominant segments—"Pay & Read" and "Wait & Read"—that together account for 72% of consumption and 96% of purchases. Validation against field experiments shows that the estimated rewards are stable and disentangled from environment dynamics. Counterfactuals yield three key insights. First, providing content for free can counterintuitively increase purchases within the consumer, by allowing them to move past likely churn points and creating additional purchase opportunities. Second, customized policies based on heterogeneous patience can increase purchases by 7%. Third, leveraging unstructured content to align pricing with narrative peaks can significantly boost monetization. More broadly, we introduce adversarial estimation as a scalable tool for dynamic models with high-dimensional states.

1 Introduction

Across a wide range of digital platforms—including streaming services (e.g., Netflix), educational platforms (e.g., Coursera), and serialized fiction apps (e.g., Wattpad)—users engage with episodic content. In these environments, consumers make a sequence of interdependent decisions: whether to continue to the next installment, pay for early access, or exit altogether. These decisions are shaped not only by structured platform features, such as pricing and access timing, but also by the *content*. A suspenseful cliffhanger may trigger immediate continuation, while a meandering or dissatisfying episode may prompt exit. Importantly, consumption is forward-looking: users weigh the immediate cost of access against the value of continuing further into the series, with content shaping how those trade-offs play out.

Understanding how content interacts with user preferences is central to platform design. Platforms must decide not only which stories to promote, extend, or discontinue, but also how to structure access—through pricing, delays, or other frictions that govern when and at what cost episodes become available. Effective personalization further requires recognizing that users differ systematically in their sensitivity to both content and access frictions. Addressing these challenges calls for a model that explains engagement and drop-off across heterogeneous users and can simulate responses to alternative release strategies, pricing policies, and targeted interventions.

To this end, we develop a framework for modeling serialized content consumption that links users' sequential decisions to both structured platform features and unstructured episode content. We cast each decision—whether and when to consume the next episode—as part of a Markov decision process (MDP), and recover a flexible reward function and a forward-looking policy that rationalize observed behavior. Our approach builds on adversarial inverse reinforcement learning (AIRL) from the computer science literature, but extend it in key ways for marketing and econometric applications. In particular, we incorporate unobserved consumer heterogeneity, allow for action-dependent utilities, and develop a scalable approach to representing unstructured content that is suitable for adversarial reward learning. Together,

these extensions enable counterfactual analysis of content pricing and release strategies.

Modeling serialized content consumption naturally fits within a dynamic discrete choice (DDC) framework, in which forward-looking users make sequential decisions based on an evolving state (Rust, 1987; Hotz and Miller, 1993; Aguirregabiria and Mira, 2002). A central challenge, however, is that the state includes not only structured variables such as prices and delays, but also unstructured content with narrative and emotional attributes, rendering the state space inherently high-dimensional. Traditional DDC models are typically tractable only in low-dimensional settings and thus struggle to scale when content plays a central role.

Existing approaches address this challenge by either distilling content into a small set of theory-driven features—such as emotional tone or narrative arc (Toubia et al., 2021; Berger et al., 2023; Knight et al., 2024)—or absorbing content effects through fixed effects (Ishihara et al., 2023; Zhao et al., 2023). However, these strategies require strong assumptions about which aspects of content matter or preclude meaningful prediction for new or out-of-sample content. More broadly, conventional DDC models rely on restrictive functional forms that limit flexibility and scalability in content-rich environments.

A parallel literature in machine learning—inverse reinforcement learning (IRL)—shares the same objective as DDC models: recovering a reward (utility) function that rationalizes observed behavior. In particular, maximum entropy IRL (Ziebart et al., 2008) models behavior as probabilistic reward maximization and accounts for both immediate and expected future value. While this framework admits flexible function approximators, likelihood-based IRL methods face computational bottlenecks analogous to those in traditional DDC estimation when the state space is high-dimensional.

We therefore build on adversarial inverse reinforcement learning (AIRL; Fu et al. 2017), which avoids explicit likelihood evaluation by adversarially aligning simulated and observed decision trajectories. Inspired by generative adversarial networks (GANs; Goodfellow et al. 2014), AIRL trains a policy to generate behavior that a discriminator—parameterized to embed a reward function—cannot distinguish from the data.

Crucially, the discriminator is constrained to decompose rewards into an immediate utility component and a continuation value, mirroring the Bellman equation. This structure helps separate preferences from environment dynamics, so the recovered immediate utility can be used for counterfactual prediction. Both the reward and policy are parameterized flexibly with neural networks, enabling tractable estimation in high-dimensional state spaces. Recent econometric work shows that adversarial estimators can recover structural primitives consistently, establishing adversarial training as a principled alternative to likelihood-based methods (e.g., Kaji et al. 2023).

We make three methodological contributions that adapt the AIRL framework of Fu et al. (2017) to the study of serialized content consumption. First, we extend AIRL to account for unobserved consumer heterogeneity. Fu et al. (2017) recovers a single reward and policy. Yet, content consumption is subjective—what compels one user to pay after a cliffhanger may leave another indifferent. We therefore estimate segment-specific reward and policy functions, jointly inferring latent segment membership. Importantly, segmentation must reflect individual behavioral regularities: a given consumer is likely to respond consistently to marketing mix and state dependence *across different series* (Ainslie and Rossi, 1998; Seetharaman et al., 1999). To capture this, we introduce a consistency constraint that links segment assignments across series consumed by the same individual.

Second, we relax the restrictive assumption that the immediate rewards do not vary directly with actions. Fu et al. (2017) restricts actions to influence rewards only through state transitions in order to guarantee that the recovered immediate reward is disentangled from environment dynamics. This assumption is untenable in our setting, where different actions (e.g., paying for early access versus waiting) naturally yield different utilities. We show that action-dependent utilities can be identified, provided that there exist an anchor action with known utilities at every state.

Third, we address how unstructured episode content should enter the state space. While pretrained large language model (LLM) embeddings provide a natural starting point, they

are optimized for semantic similarity rather than decision relevance and encode substantial variation—such as stylistic details or named entities—that is orthogonal to consumption incentives. We therefore adopt a two-stage design in which episode content is first compressed using a corpus-level autoencoder, and preferences over this representation are then inferred through AIRL. This separation denoises content while preserving narrative structure, allowing the recovered reward to reflect how users value content and enabling counterfactual analysis for unseen episodes.

More broadly, we introduce adversarial estimation to the marketing literature. Our heterogeneous adversarial framework extends beyond content consumption and can be applied to static models or to settings with rich structured covariates, opening new opportunities for structural analysis in high-dimensional settings.

We apply this framework to a large-scale dataset from a serialized fiction platform that employs a "wait-for-free" (WFF) pricing policy—a form of temporal versioning in which episodes can be unlocked immediately via payment or accessed for free after a fixed delay. The data span 15 months and cover over one million users and 10,000 serialized titles, with detailed episode-level records indicating whether access occurred via payment or waiting. While our application focuses on textual content in serialized fiction, the framework generalizes to other media—such as audio, video, or interactive content—where engagement unfolds through sequential, content-dependent choices.

The model estimates reveal pronounced heterogeneity in willingness to pay, willingness to wait, and sensitivity to content. Two user segments dominate engagement: the *Pay & Read* segment (12% of users) and the *Wait & Read* segment (23%). Both progress far into series, but differ sharply in monetization behavior—*Pay & Read* segment frequently pays for immediate access, while *Wait & Read* segment relies primarily on delayed free access. Longer wait times increase purchase likelihood for both segments, but also raise exit probabilities, with the exit response more than twice as large for the *Wait & Read* segment. This asymmetry highlights a fundamental trade-off between monetization and retention:

delays induce payment, but disproportionately push delay-sensitive users to disengage.

The segments also differ sharply in how content shapes decisions. *Pay & Read* segment responds strongly to cliffhangers and faster-paced episodes, which shift behavior toward immediate purchase rather than waiting or exit. In contrast, the *Wait & Read* segment exhibits relatively muted responses to content variation. Overall, consumption decisions for *Wait & Read* segment is driven primarily by access frictions rather than narrative appeal.

We validate the model using field experiment data from an exogenous platform intervention that altered wait-times for a subset of books excluded from estimation (Todd and Wolpin, 2006; Misra and Nair, 2011). The model accurately predicts segment-level changes in consumption following the policy shift, providing rare out-of-sample validation for a forward-looking structural model. This evidence indicates that the recovered reward functions reflect stable, intrinsic preferences, supporting their use for counterfactual policy analysis.

Our first counterfactual evaluates the impact of introducing WFF. Contrary to standard freemium logic, which predicts purchase cannibalization, we find that WFF increases both consumption and purchases within the *Pay & Read* segment—raising total consumption by 52% and purchases by 36%. Free access allows these users to move past weaker episodes that might otherwise trigger exit, creating additional monetization opportunities.

Building on these results, our second counterfactual examines how platforms can reduce cannibalization risk while preserving the monetization benefits of WFF. Because users differ in both willingness to pay and tolerance for delay, we tailor wait-times by segment: 24-hour delay for *Wait & Read* segment and 1-hour delay for *Pay & Read* segment. Longer delays reduce engagement by inducing early exit among low-patience, low-WTP users, while increasing purchase propensity among those who remain, shifting consumption toward paid access. As a result, total purchases increase by 7% despite a 17% decline in overall consumption.

Finally, we ask whether content itself can serve as a pricing lever. Serialized media evolves over time: some episodes are low-stakes fillers, while others end on high-suspense cliffhangers. Can platforms design access rules that adapt to this variation, rather than a one-size-fits-all

policy? We evaluate a content-based policy that designates high-engagement episodes as paid-only while retaining WFF access for weaker episodes. This targeted approach substantially increases monetization, especially for the *Wait & Read* segment—raising purchases by up to 225%—while leaving overall consumption essentially unchanged.

The remainder of the paper is organized as follows. Section 2 reviews the related literature. Section 3 presents the institutional context and data. Sections 4 and 5 describe the methodological background and model. Section 6 reports the estimation results, and Section 7 presents counterfactual analyses. Section 8 concludes.

2 Related Literature

2.1 Consumption of Serialized Content

We contribute to the growing literature on serialized content consumption, which studies how partitioned formats—such as chapters or episodes—shape engagement and interact with monetization strategies such as paywalls and versioning. Prior work documents distinctive behavioral patterns, including binge consumption, and their implications for advertising effectiveness, retention, and subscription behavior (Schweidel and Moe, 2016; Godinho de Matos and Ferreira, 2020; Zhang et al., 2022). More broadly, this work relates to classic theories of digital content monetization that examine how hybrid access rules—combining free sampling with paid consumption—shape demand and revenue (Halbheer et al., 2014).

Firms increasingly tailor monetization strategies to the sequential structure of serialized content. Zhao et al. (2023) show that combining simultaneous and staggered release models can increase platform revenue, while Ishihara et al. (2023) find that flexible viewing schedules raise ratings and advertising exposure. A particularly relevant approach is the "wait-for-free" (WFF) model, in which users either pay for immediate access or wait to consume an episode for free after a fixed delay. Recent studies (Choi et al., 2025; Lee et al., 2025) show that

wait-time length plays a central role in shaping both engagement and monetization.[1]

Our work also connects to Fong and Gui (2024), who use large language models (LLMs) to generate hypothetical continuations of serialized narratives and quantify reader expectations based on semantic dimensions such as emotion, surprise, and psychological themes.

We extend this literature by developing a dynamic model of serialized content consumption that incorporates high-dimensional content representations as part of the evolving state space. The model captures how content influences forward-looking decisions and allows for latent user heterogeneity. By moving beyond static predictors and coarse summaries, our framework enables platforms to assess demand for new content and counterfactual policies.

2.2 Inverse Reinforcement Learning

We build on the inverse reinforcement learning (IRL) literature from computer science, which provides a parallel framework to dynamic discrete choice (DDC) models in economics. Foundational work by Ng and Russell (2000) and Abbeel and Ng (2004) formalized IRL as recovering a reward function and an associated policy from *expert demonstrations*, under the assumption that observed behavior is (approximately) optimal.

A leading approach is maximum entropy (MaxEnt) IRL (Ziebart et al., 2008), which selects the highest-entropy policy among those consistent with the demonstrations. Under T1EV errors, MaxEnt IRL is mathematically equivalent to logit-based DDC models (Haarnoja et al., 2018; Sanghvi et al., 2021; Cen et al., 2022), linking reward recovery in IRL to utility estimation in structural choice models. A key distinction lies in the source of randomness: DDC models attribute it to structural errors, while MaxEnt IRL treats it as intrinsic to the agent's stochastic policy.

Both traditional DDC and MaxEnt IRL face severe computational challenges in high-dimensional state spaces. DDC requires repeated evaluation of continuation values via in-

[1]Modeling content consumption under WFF relates to work on varying content availability times (e.g., Luan and Sudhir 2022) and to broader approaches that model consumption dynamics when usage is observed (e.g., Huang et al. 2015; Chou and Kumar 2024).

tegration over next states, which quickly becomes intractable as the state dimension grows. Although this has motivated approximation and simulation-based methods (e.g., Hotz et al. 1994; Keane and Wolpin 1994; Aguirregabiria and Mira 2002), estimation remains difficult when the state space is rich because transitions must be represented and learned from sparse data, leading to finite-sample bias. MaxEnt IRL faces an analogous bottleneck as trajectory likelihood evaluation also entails integrating over future states and transition dynamics.

As a result, solving IRL via maximum likelihood remains an open challenge (Zeng et al., 2023, 2025). Recent work by Kang et al. (2025) addresses this issue by leveraging an "anchor action" with known rewards. This enables joint minimization of the negative log-likelihood and a Bellman error term evaluated only at the anchor actions. This bypasses explicit transition estimation, allowing for scalable inference with provable convergence guarantees.

An alternative is adversarial IRL, which replaces likelihood evaluation with an adversarial game that matches observed and simulated trajectory distributions (Finn et al., 2016; Fu et al., 2017). A policy generates trajectories while a discriminator learns to distinguish them from observed behavior, enabling scalable reward learning in complex environments. Additional background is provided in Section 4 and Online Appendix Section W1.

Beyond their empirical success, adversarial methods also have growing theoretical support in econometrics and statistics. Kaji et al. (2023) establish conditions under which adversarial estimators consistently recover structural primitives in simulation-based settings, and Puchkin et al. (2024) characterize convergence rates for GAN-type estimators in nonparametric problems. Together, these results position adversarial training as a statistically principled alternative to likelihood-based estimation in high-dimensional environments.

More broadly, the use of neural networks for estimating structural models is an active area of research. Wei and Jiang (2025) propose a neural net estimator that learns the inverse mapping from simulated moments to parameters, offering a computationally light alternative to simulated method of moments. Their focus is on reducing the simulation burden, whereas we focus on solving high-dimensional dynamic problems through an adversarial approach.

We develop an adversarial IRL framework for serialized content consumption by extending Fu et al. (2017) to allow for unobserved heterogeneity. AIRL rationalizes behavior with a single reward and policy, but in our setting users differ systematically in their preferences (e.g., willingness-to-pay, tolerance for delay). We address this by estimating segment-specific reward and policy functions and jointly inferring latent segment membership. Although prior imitation learning work considers multiple policies (Li et al., 2017; Fu et al., 2023), AIRL has not been adapted to recover heterogeneous preferences.

3 Institutional Details and Problem Setup

3.1 Empirical Context

Our empirical setting is a U.S.-based serialized fiction platform that features romance content. The platform hosts over 10,000 series (i.e., books), each divided into sequential episodes, and serves more than one million active users. Content is accessed exclusively via a mobile application, where users make consumption decisions one episode at a time. Figure W2.1 provides a representative illustration of the app interface.

The platform employs a wait-for-free (WFF) monetization policy, a form of temporal versioning. Initial episodes of each series are freely accessible. Once a user reaches later episodes, access is governed by a time-based constraint: users may either pay a fixed price (∼$0.50) to unlock the next episode immediately or wait a predetermined period to access it for free. This pay-versus-wait tradeoff applies sequentially as users progress through the series and accounts for the majority of consumption and purchases on the platform.

Wait-times are fixed at the series level—typically ranging from 1 to 72 hours—and apply uniformly across users and episodes. All series follow a forward-only pricing structure, beginning with free episodes, followed by WFF episodes, and in some cases concluding with paid-only content. Prices do not vary at the episode level. This institutional structure limits the scope for price endogeneity and simplifies the consumer's decision problem to a choice

among immediate access, delayed access, or exit. Related discussions of the WFF mechanism and its behavioral implications can be found in Choi et al. (2025) and Lee et al. (2025).

3.2 Data

Our analysis draws on a comprehensive dataset, which includes three key components:

1. **Consumption Panel:** A user-episode panel over a 15-month period (October 1, 2020 to December 1, 2021). For each consumption event, we observe the timestamp, access mode (purchase vs. wait), and the corresponding user, series, and episode identifiers.

2. **Series and Episode Metadata:** Structured attributes for each series and episode, including title, author, episode sequence, pricing, release date, and required wait-time.

3. **Episode Text:** Full textual content for all episodes in the first season of each series.

To construct a tractable dataset for IRL training, we apply a set of filters. We restrict attention to WFF series that satisfy the following criteria: (i) $\geq$ 1,000 unique readers; (ii) wait-time $\leq$ 24 hours; (iii) $\leq$ 100 episodes in the first season; (iv) fully published prior to the start of the observation window; and (v) no wait-time changes during the study period. These restrictions ensure that each series has sufficient volume and a stable consumption environment. We exclude user-series trajectories with irregular behavior—e.g., skipping or reading episodes out of sequence—which are rare (less than 5% of observations), and retain only the users who read at least five distinct series from the selected pool.

The final estimation sample consists of approximately 290,000 user–series trajectories from 24,000 users across 151 series and 6,002 episodes. Summary statistics are reported in Table 1. We split the data at the series level, using 80% of series for training and reserving the remaining 20% for out-of-sample evaluation. This design ensures that model validation is conducted on entirely unseen content.

	Mean	**SD**	**25%**	**50%**	**75%**
Episodes per series	40	20	25	36	49
Series consumed per user	12	7	7	9	14
Episodes consumed per user	160	142	64	122	211
Episodes purchased per user	67	72	25	46	84
Days spent per user per series	10	32	1	1	5

Table 1: Summary statistics

Field Experiment Dataset To assess the validity of our model, we leverage a field experiment that enables out-of-sample validation using consumer responses to an exogenous platform intervention. Accurately predicting the behavioral impact of this intervention provides evidence that the model has learned stable preferences—strengthening the credibility of its counterfactual analysis under alternative platform policies.

Beginning in November 2020, the platform implemented permanent wait-time reductions for a subset of WFF series through a staggered rollout. These changes were applied at the series level and varied in magnitude. For instance, a series that previously required a 24-hour wait between episodes was updated to allow free access every hour instead. As a result, the intervention introduced exogenous variation in episode access timing across series. Additional details on the rollout of the wait-time reduction are provided in Lee et al. (2025).

Applying the same sample restrictions yields a field experiment dataset of approximately 51,000 user–series trajectories from 17,000 users, spanning 93 books and 4,498 episodes. From this, we construct two disjoint subsets: one under the original, longer wait-times (pre-reduction), and the other after the policy change (post-reduction). These series are not used for model training, allowing us to assess whether the estimated model accurately predicts behavioral shifts in response to exogenous variation in wait-time policy.

Reduced-Form Evidence of Forward-Looking Behavior Consumption of serialized content is inherently forward-looking: users form expectations about the utility of future episodes, and these expectations shape current decisions. Prior work documents such dynamics in books, video games, and live entertainment (Simonov et al., 2023; Fong and Gui,

2024; Knight et al., 2024). We provide direct evidence of forward-looking behavior in our setting using variation in the platform's wait-for-free (WFF) policy.

Under WFF, the wait-time only matters once users reach the paywall, but it is known from the start of a series. This creates a useful exclusion restriction: wait-time shifts the expected value of continuing while leaving the utility of the current free episode unchanged. If consumers are forward-looking, continuation during free episodes should already respond to the wait-time governing later access.

We test this prediction by estimating a logistic regression of continuation from one free episode to the next. The specification includes book fixed effects, episode-level content controls, and interactions between the series-level wait-time and a proxy for willingness to pay.[2] Users are classified as *spenders* or *non-spenders* based on prior platform spending, with the latter more reliant on delayed free access.[3]

Table 2 reports the results. The coefficient on the non-spender indicator is negative (-0.076), indicating that non-spenders are less likely to continue on average. Importantly, the interaction between wait-time and the non-spender indicator is also negative (-0.005): for users who rely on free access, longer future waits reduce the likelihood of continuation even when the next episode is free.

Turning to content, one key episode-level feature is cliffhanger strength, measured using narrative elements such as surprise and suspense (Online Appendix Section W5). The positive coefficient on Cliffhanger (0.059) shows that stronger cliffhangers increase continuation. However, its interaction with wait-time is negative (-0.009), suggesting that longer future waits dampen this otherwise propelling effect. These patterns together provide evidence of forward-looking behavior: consumers weigh not only the appeal of the current episode but also the expected utility of continuing further into the series.

[2]Episode-level content features include cliffhanger strength, narrative dynamics, and emotional tone, which are extracted from the full episode text and described in detail in Section 5.1.

[3]While we do not observe the specific content of purchases prior to the panel period, we observe total spending amounts. This binary classification serves as a coarse proxy for users' willingness to pay.

DV: Pr(Continue)	Coefficient	Std. Error
Intercept	2.171***	(0.071)
$\mathbb{1}$(Non-spender)	-0.076***	(0.018)
$\mathbb{1}$(Non-spender) × Wait-time	-0.005***	(0.001)
Cliffhanger	0.059***	(0.004)
Cliffhanger × Wait-time	-0.009***	(0.001)
Episode Content Features	Y	
Book FE	Y	
N Obs	1,296,329	
N Books / Episodes	151 / 842	

* $p < .1$, ** $p < .05$, *** $p < .01$

Table 2: Logistic regression of continuation from one free episode to the next. Controls: book FE, episode-level content features. Standard errors clustered at the user level.

Unobserved Heterogeneity in Consumption Patterns We conclude this section by documenting empirical patterns that motivate explicit modeling of consumer heterogeneity. Figure 1 plots the distribution of two user-level outcomes across series: completion rate (the fraction of a series read; left panel) and purchase probability (purchases divided by episodes consumed; right panel). Both exhibit substantial dispersion. Some users exit after only a few episodes, while others routinely complete entire series. Similarly, purchase behavior ranges from users who almost exclusively wait for free access to those who frequently pay to unlock episodes. These distributions point to systematic differences in willingness to pay and tolerance for delay that cannot be captured by a single representative consumer.

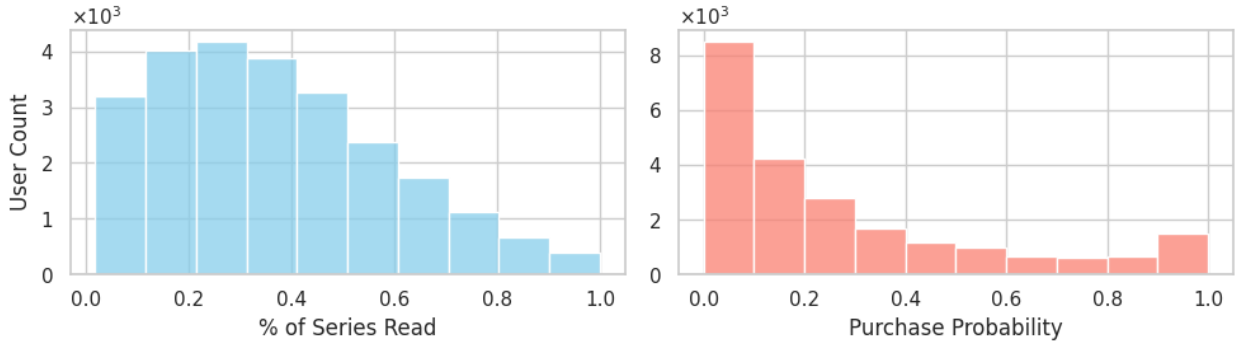

Figure 1: Distribution of completion rate (left) and purchase probability (right) across users

Beyond heterogeneity across users, we find strong evidence that preferences are correlated within users across series, underscoring the need to model persistent user-level structure. Ta-

ble 3 reports two complementary statistics: (i) the average pairwise correlation in outcomes across series within a user, and (ii) the share of variation explained by user fixed effects. Purchase behavior is highly persistent, with a within-user correlation of 0.50 and user fixed effects explaining 61% of the variation in purchase probabilities. Completion rates also exhibit meaningful persistence, with a within-user correlation of 0.18 and 29% of the variance explained by user fixed effects.

	% of Series Read	**Purchase Probability**
Within-user avg. correlation	0.176	0.500
User fixed effect R^2	0.288	0.608

Table 3: Number of series by wait-time (in hours) for pre- and post-reduction

Together, these patterns indicate that users differ systematically in their consumption preferences and that these differences persist across product categories (Ainslie and Rossi, 1998; Seetharaman et al., 1999). This motivates a modeling framework that accommodates both heterogeneity across users and correlation in preferences within users.

3.3 Problem Formulation

We model user consumption as an infinite-horizon MDP $\mathcal{M} = \{\mathcal{S}, \mathcal{A}, \mathcal{T}, \mathcal{R}, \gamma\}$, where $\mathcal{S}$ denotes the state space, $\mathcal{A}$ the action set, $\mathcal{T}$ the transition function, $\mathcal{R}$ the reward function, and $\gamma \in (0, 1)$ the discount factor.[4] For each series j, user i enters the decision process upon consuming the first episode and proceeds episode by episode. At each decision step t—indexed by episode sequence—the user selects an action $a_{ijt} \in \mathcal{A} = \{0, 1, 2\}$, corresponding to (a) immediate consumption, (b) delayed consumption, or (c) exit.[5] For WFF episodes, immediate consumption requires payment, while delayed consumption entails waiting for free access.

The state vector $s_{ijt} \in \mathcal{S}$ summarizes the user's information set at decision step t in

[4]It is well known in the literature that the discount factor cannot be separately identified from rewards with choice data alone. We fix the discount factor at $\gamma = 0.99$, applied per episode. Following the inverse reinforcement learning literature, we refer to instantaneous utility as a *reward*, $r(s, a)$.

[5]Time t is indexed by episode rather than calendar time, consistent with prior work on dynamic content consumption (Zhao et al., 2022; Amano and Simonov, 2024; Lee et al., 2025).

series j and includes: (1) $\mathcal{C}_{ijt}$: content representation of the preceding episode, (2) g_{ijt}: user progress (fraction of episodes consumed), (3) $\mathbf{P}^0_{ijt}, \mathbf{P}^1_{ijt}$: price indicators for immediate and delayed access for the current and next episode, (4) w_j: series-specific wait-time for free access, (5) $pay_{ij(t-1)}$, indicator for whether the previous episode was purchased, and (6) $\mathbf{X}_{ijt}$: hand-crafted narrative features of the previous episode. Details on the content representation and narrative features are later provided in Section 5.1. State transitions are deterministic conditional on the structure of series j.

A consumption trajectory for user i in series j is denoted as $\tau_{ij} = (s_{ij0}, a_{ij0}, s_{ij1}, a_{ij1}, \dots)$ and terminates either when the user exits ($a_{ijt} = 2$) or completes the final episode ($g_{ijt} = 1$). In both cases, the user enters the absorbing state with zero rewards.

To capture unobserved heterogeneity, we assume each user belongs to a latent segment $z \in \{1, \dots, K\}$, where K is pre-specified. Users within a segment share a common reward function $r^*(s, a, z)$ and policy $\pi^*(a \mid s, z)$. The policy is a stochastic mapping from states to actions, analogous to conditional choice probabilities in dynamic discrete choice models.

Following the maximum-entropy IRL framework, we assume users stochastically maximize expected discounted utility:

$$\pi^*(a \mid s, z) = \arg\max_{\pi} \; \mathbb{E}_{\tau \sim \pi}\left[\sum_{t=0}^{\infty} \gamma^t r^*(s_t, a_t, z) + \mathcal{H}(\pi)\right], \tag{1}$$

where $\mathcal{H}(\pi)$ denotes policy entropy. Our objective is to recover the segment-specific reward functions $r^*(s, a, z)$ and corresponding policies $\pi^*(a \mid s, z)$ that best rationalize observed consumption trajectories.

4 Methodological Background

To estimate reward functions in high-dimensional state spaces, a growing class of inverse reinforcement learning methods adopts an adversarial perspective. Drawing inspiration from GANs (Goodfellow et al., 2014), these approaches treat the policy as a generator that sim-

ulates trajectories intended to match observed behavior, while a discriminator learns to distinguish observed from policy-generated transitions (Finn et al., 2016; Fu et al., 2017). This setup can be viewed as minimizing a divergence between the empirical trajectory distribution and the distribution induced by the policy. Whereas maximum likelihood estimation corresponds to minimizing Kullback–Leibler divergence, adversarial training allows optimization over a broader class of divergences (Grover et al., 2018; Gao et al., 2018).

Among these approaches, AIRL (Fu et al., 2017) introduces additional structure to ensure that the recovered rewards are consistent with rational, forward-looking decision making. The discriminator is parameterized to reflect a Bellman-style decomposition, allowing the learned reward to be interpreted as the agent's instantaneous utility. The policy is then trained to maximize these rewards through standard reinforcement learning. At convergence, the discriminator can no longer distinguish observed from generated behavior, and the learned reward function rationalizes observed choices while the policy represents the optimal response under that reward.

Mathematical Formulation. The discriminator is structured to embed a Bellman consistent decomposition of rewards and values, allowing the recovered reward to be interpreted as the agent's instantaneous utility. Formally, the discriminator classifies each state–action–next-state triplet (s, a, s') according to

$$D_{\theta,\phi}(s, a, s') = \frac{\exp\{f_{\theta,\phi}(s, a, s')\}}{\exp\{f_{\theta,\phi}(s, a, s')\} + \pi(a \mid s)} \tag{2}$$

$$\text{where} \quad f_{\theta,\phi}(s, a, s') = g_\theta(s, a) + \gamma h_\phi(s') - h_\phi(s) \tag{3}$$

Here, $\pi(a \mid s)$ denotes the probability of taking action a in state s under the current policy, which is held fixed during discriminator training. The function $f_{\theta,\phi}(s, a, s')$ decomposes into an immediate reward component $g_\theta(s, a)$ and a continuation value adjustment $\gamma h_\phi(s') - h_\phi(s)$. This structure corresponds to a one-step temporal-difference update and can be interpreted as a soft Bellman residual, where $h_\phi(s)$ approximates the value function—or

inclusive value under T1EV assumptions. The discriminator therefore updates rewards in a manner consistent with forward-looking decision making.

The discriminator can be interpreted as estimating the relative likelihood that a transition originates from the observed data rather than the policy-induced distribution:

$$D(s, a, s') = \frac{P_{data}(s, a, s')}{P_{data}(s, a, s') + P_{policy}(s, a, s')}.$$

Given the discriminator, AIRL defines a shaped reward for each transition as

$$\tilde{r}(s, a, s') = \log D_{\theta,\phi}(s, a, s') - \log(1 - D_{\theta,\phi}(s, a, s')) = f_{\theta,\phi}(s, a, s') - \log \pi(a \mid s)$$

The policy is then updated to maximize expected cumulative shaped rewards, yielding the entropy-regularized objective

$$\max_{\pi} \ \mathbb{E}_{\pi}\left[\sum_{t=0}^{T} f_{\theta,\phi}(s_t, a_t, s'_t)\right] + \mathcal{H}(\pi),$$

which corresponds to the maximum-entropy RL problem in Equation 1. Policy optimization is carried out using standard gradient-based policy optimization algorithms such as Proximal Policy Optimization (PPO; Schulman et al. 2017).

At convergence, the discriminator is unable to distinguish observed from generated transitions, implying $D = 1/2$ and

$$\begin{aligned} f^*(s, a, s') &= \log \pi^*(a \mid s) = Q^*(s, a) - V^*(s) \\ &= r^*(s, a) + \gamma \mathbb{E}_{s'}[V^*(s')] - V^*(s) \end{aligned}$$

Consequently, $g_\theta(s, a)$ and $h_\phi(s)$ recover the agent's immediate reward and continuation value, respectively, up to an additive constant.[6]

[6] In dynamic discrete choice models, the continuation value $\mathbb{E}_{s'}[V(s')]$ is evaluated by integrating over the known transition matrix $T(s' \mid s, a)$. Under rational expectations, the agent's subjective transition beliefs coincide with T. In AIRL, this expectation is evaluated by Monte Carlo: the simulator generates next states $s' \sim T(\cdot \mid s, a)$ along sampled trajectories, and averages of $V(s')$ across these draws approximate the same continuation value without requiring analytic integration.

Action-dependent Utilities The theoretical guarantee in Fu et al. (2017) establishes identification of rewards only under the restriction that the reward function is state-dependent, $g(s)$. If rewards depend on actions, $g(s, a)$, then the decomposition of $f(s, a, s') = g(s, a) + \gamma\, h(s') - h(s)$ is not unique: variation in g can be offset by changes in h. Intuitively, when an action leads to favorable future states, it is ambiguous whether the observed choice is driven by a high immediate payoff or by continuation value. Without additional structure this leads to non-identification, and Fu et al. (2017) resolve the problem by restricting to state-only rewards, which are then identified up to a constant shift.

In our setting, however, the environment itself provides a natural normalization. At every state, the consumer has an *exit option* with known flow payoff of zero, $g(s, a = exit) = 0$, which leads deterministically to an absorbing state with zero continuation value, $h(s_{abs}) = 0$. These anchors eliminate the ambiguity in the (g, h) decomposition and uniquely pin down action dependent rewards, $g^*(s, a) = r^*(s, a)$. The is consistent with the normalizing assumption required for identification in the DDC literature Rust (1994); Magnac and Thesmar (2002). We provide formal proof in Online Appendix Section W3.

5 IRL for Modeling Serialized Content Consumption

Figure 2 illustrates the overall structure and training workflow of our inverse reinforcement learning framework. The model is trained iteratively using the following sequence:

S0: MDP Setup and Network Initialization. Define the MDP environment and initialize the policy, discriminator (reward), and posterior networks.

S1: Trajectory Simulation. Simulate trajectories by allowing the policy to interact with the environment, conditioned on randomly sampled series and latent segments.

S2: Discriminator Update. Update the discriminator (reward function) by minimizing classification loss in distinguishing observed from simulated transitions.

S3: Posterior Update. Train the posterior network to infer latent segments using classification and within-user consistency losses.

S4: Reward Assignment. Assign rewards to simulated trajectories based on the updated discriminator and posterior.

S5: Policy Update. Update the policy using reinforcement learning to maximize expected cumulative reward.

Steps S1 through S5 are repeated until convergence, jointly refining the reward function and policy while accounting for unobserved user heterogeneity.

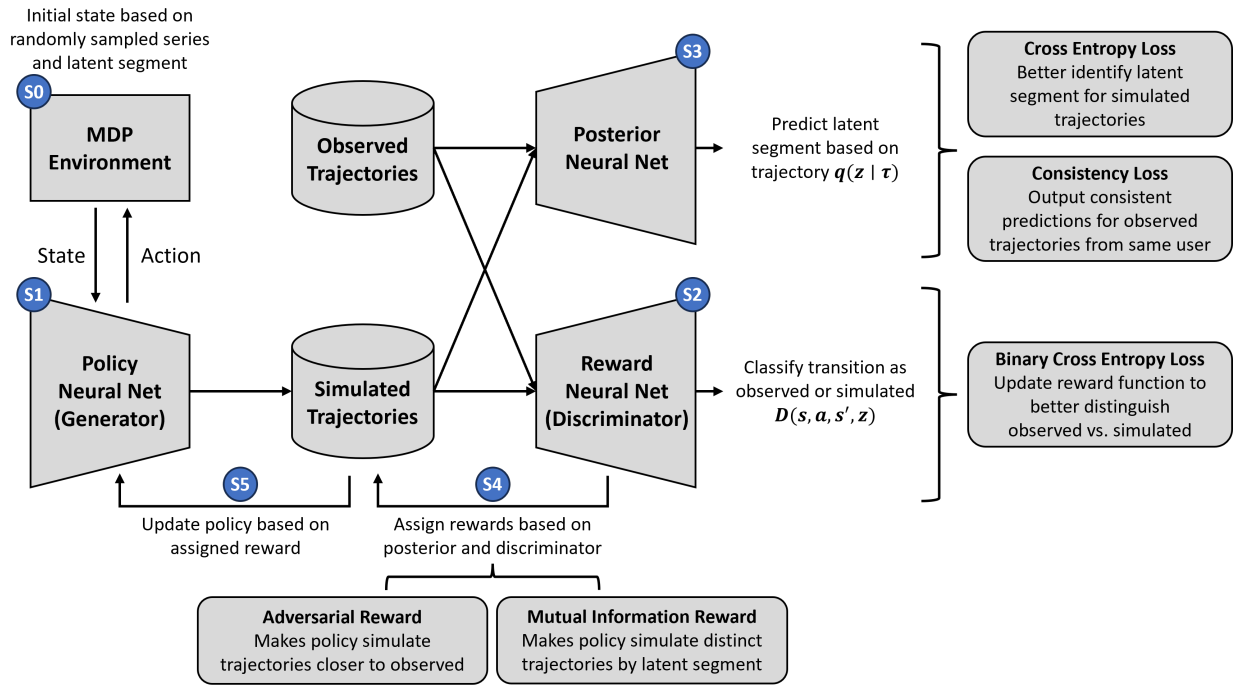

Figure 2: IRL Algorithm for Serialized Content Consumption

5.1 S0. MDP Setup and Network Initialization

We formalize serialized content consumption as a Markov decision process (MDP) and initialize the core components of the learning architecture: the policy, the discriminator (reward function), and the posterior network for latent segment inference. A central challenge in

defining the MDP is how to represent *unstructured textual content* in the state space in a way that supports generalization across series with highly heterogeneous narratives.

Unlike standard IRL applications in robotics or navigation—where unstructured inputs typically consist of images processed by convolutional neural networks—our setting features natural language text. Each episode contains narrative content that can meaningfully influence user decisions, yet also varies widely across books in style, vocabulary, and genre.

A natural starting point is to use off-the-shelf embeddings from large language models (LLMs) as state inputs. While such representations capture rich semantic information, they also encode high-variance, series-specific features—such as named entities, stylistic markers, or genre-specific terminology—that are largely orthogonal to consumption decisions (Bommasani et al., 2021). As a result, pretrained embeddings can allocate a substantial share of representational variance to series identity even when episodes elicit very different behavioral responses. In IRL settings, this variance structure can dominate the state space and allow the discriminator to exploit idiosyncratic, non-transferable patterns, rather than learning reward signals that generalize across narratives.

To address this challenge, we project high-dimensional LLM embeddings onto a lower-dimensional latent space using an autoencoder trained on a large corpus of narrative content. The autoencoder is trained to reconstruct the original embeddings—preserving semantic content—while compressing the representation to emphasize variation common across many books and downweight idiosyncratic, series-specific features. Importantly, the autoencoder is trained on a substantially larger corpus than the trajectories used for policy learning, allowing it to learn the global geometry of narrative variation at scale.

This dimensionality reduction plays a central role in stabilizing reward learning. By organizing the state space around narrative dimensions that systematically influence user decisions and recur across books, the resulting representation is smoother and more generalizable: small changes in content induce small changes in the embedding, and similar narrative patterns across series are mapped to nearby states. This structure limits the discriminator's

ability to exploit incidental variation and supports the recovery of reward functions that generalize across heterogeneous narratives.

We train the autoencoder on 437,310 episodes from 14,821 series. Each episode's text is first embedded into a 3,072-dimensional vector using OpenAI's `text-embedding-3-large` model, which the encoder compresses into a 32-dimensional latent representation. These embeddings are then used as part of the state space for policy and reward learning. Empirically, we find that this representation enables stable convergence of the AIRL algorithm, whereas directly using pretrained embeddings—either at full dimension or truncated via off-the-shelf methods—leads to unstable training and fails to converge.

Figure 3 visualizes the variance structure of alternative content representations using two-dimensional PCA projections. Pretrained LLM embeddings—both at full dimension and truncated via Matryoshka representation learning—exhibit diffuse variance across multiple directions, indicating that a large share of representational capacity is allocated to high-variance, series-specific features. By contrast, the autoencoder embeddings concentrate variance into a more compact subspace, yielding a smoother representation in which dominant directions reflect narrative dimensions that recur across books.

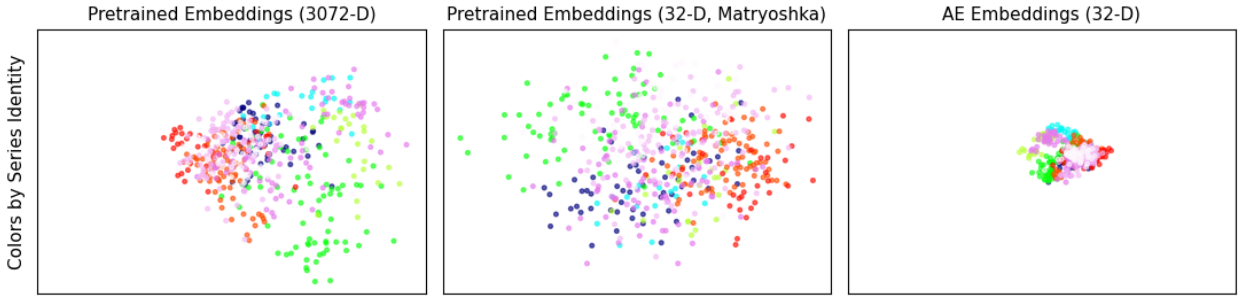

Figure 3: Top 2 principal components for episode embeddings drawn from randomly sampled series. Left: pretrained LLM embeddings (3072-D). Middle: pretrained LLM embeddings truncated to 32 dimensions via Matryoshka representation learning. Right: 32-D embeddings learned via a corpus-scale autoencoder. Colors indicate series identity.

To complement content representations, we also extract a set of theory-driven narrative features linked to content consumption. Specifically, we measure *cliffhanger strength*, capturing the extent to which an episode's ending motivates continued reading (Lee et al., 2025);

narrative dynamics, including speed (pacing), volume (semantic breadth), and circuitousness (narrative directness) (Toubia et al., 2021); and *emotional tone*, measured through valence (positivity) and arousal (emotional intensity) (Mendes and Martins, 2023). Additional details on feature construction are provided in Online Appendix Section W5.

5.2 S1. Policy and Environment: Simulating Trajectories

To simulate trajectories, the stochastic policy interacts with the MDP environment that emulates user consumption of serialized content. At the beginning of each trajectory, a series j and a latent segment $z \in \{1, \ldots, K\}$ are sampled uniformly at random, which remain fixed throughout the trajectory. At each time step t, the environment provides the current state s_t. Given s_t and z, the policy outputs a probability distribution over discrete actions, denoted by $\pi(a_t \mid s_t, z)$. Action a_t is sampled stochastically and the environment then updates the state based on the selected action and the structure of the sampled series.

Each trajectory terminates either when the user chooses to exit or reaches the final episode of the series, after which an absorbing state is appended. These simulated trajectories are retained for the current training iteration and used to update the policy, reward function, and latent segmentation. New trajectories are then generated in the next training cycle.

5.3 S2. Discriminator Network Training

The discriminator updates a segment-specific reward function by distinguishing observed transitions from those generated by the current policy. As in Section 4, the discriminator is parameterized to enforce a Bellman-consistent decomposition, with all components now conditioned on the latent segment z. Specifically, for each transition (s, a, s') we define

$$D_{\theta,\phi}(s, a, s', z) = \frac{\exp\{f_{\theta,\phi}(s, a, s', z)\}}{\exp\{f_{\theta,\phi}(s, a, s', z)\} + \pi(a \mid s, z)} \tag{4}$$

where $\pi(a \mid s, z)$ is the current policy (held fixed during discriminator training), and

$$f_{\theta,\phi}(s, a, s', z) = g_\theta(s, a, z) + \gamma h_\phi(s', z) - h_\phi(s, z). \tag{5}$$

The term $g_\theta(s, a, z)$ represents instantaneous utility for segment z, and $h_\phi(\cdot, z)$ captures the associated continuation value.

The discriminator is trained by binary cross-entropy to classify transitions from the observed data $\mathcal{D}_{\text{obs}}$ versus simulated rollouts $\mathcal{D}_{\text{sim}}$:

$$\mathcal{L}_{\text{disc}} = \mathbb{E}_{(s,a,s')\sim\mathcal{D}_{\text{obs}}} \left[\log D_{\theta,\phi}(s, a, s', \hat{z})\right] + \mathbb{E}_{(s,a,s',z)\sim\mathcal{D}_{\text{sim}}} \left[\log\left(1 - D_{\theta,\phi}(s, a, s', z)\right)\right]. \tag{6}$$

For $\mathcal{D}_{\text{obs}}$, the latent segment z is unobserved, and hence we use $\hat{z}$ estimated through the posterior network (Step S3). In $\mathcal{D}_{\text{sim}}$, z is known by construction.

At convergence, simulated and observed transitions are indistinguishable and $D_{\theta,\phi}(s, a, s', z) \to 1/2$, implying

$$f^*(s, a, s', z) = \log \pi^*(a \mid s, z) = r^*(s, a, z) + \gamma V^*(s', z) - V^*(s, z).$$

Consequently, $g_\theta(s, a, z)$ recovers segment-specific instantaneous utility and $h_\phi(s, z)$ recovers the corresponding continuation value.

5.4 S3. Posterior Network Training

Standard AIRL recovers a single reward function and policy. While this is appropriate in canonical applications such as navigation or robotic control, it is ill-suited to settings like content consumption, where users exhibit substantial and persistent heterogeneity in preferences. As shown in Section 3.2, behavior varies widely across users yet remains internally consistent across series, implying stable latent preferences. Capturing such behavior requires jointly recovering segment-specific reward and policy functions while enforcing coherence in preferences across trajectories generated by the same user.

5.4.1 Accounting for Unobserved Heterogeneity across Users

To model unobserved heterogeneity, we augment AIRL with a discrete latent segment variable $z \in \{1, \ldots, K\}$. Both the policy and reward functions are conditioned on z, allowing behavior to vary systematically across segments. During simulation (Step S1), each trajectory is generated conditional on a fixed segment.

Because segment labels are unobserved for real user trajectories, we introduce a posterior network $q_\psi(z \mid \tau)$ that maps a full trajectory τ to a distribution over segments. This variational approximation replaces the unknown posterior $p(z \mid \tau)$ when updating the discriminator and policy, enabling joint learning of segment-specific rewards and policies.

To ensure that segments capture behaviorally distinct preferences, we encourage trajectories generated under different segments to be informative about z. Following prior work (Chen et al., 2016; Li et al., 2017), we maximize a variational lower bound on the mutual information between segments and trajectories,

$$L_{\mathrm{MI}}(\pi, q) = \mathbb{E}_{z \sim p(z),\, \tau \sim \pi(\cdot \mid z)} \big[\log q_\psi(z \mid \tau) \big] + \mathcal{H}(z),$$

where $p(z)$ is uniform. In practice, the posterior network is trained by minimizing the corresponding cross-entropy loss,

$$\mathcal{L}_{\mathrm{CE}} = \mathbb{E}_{\tau \sim \pi} \big[-\log q_\psi(z \mid \tau) \big].$$

We implement q_ψ as a Long Short-Term Memory (LSTM) network, though the approach is modular and does not depend on a specific architecture (e.g., GRUs or transformers).

Related work has used mutual information objectives to uncover latent structure from static observations or per-transition data—for example, digit identity in GANs (Chen et al., 2016) or latent policies in imitation learning (Li et al., 2017). Our setting differs fundamentally in that economically meaningful heterogeneity is revealed only through sequences of decisions. Attributes such as patience or price sensitivity are not identifiable from individual actions in isolation, particularly when narrative content evolves exogenously. We find that per-transition classifiers tend to collapse to degenerate segmentations driven by superficial

cues (e.g., action type). Conditioning inference on full trajectories via a sequential posterior is therefore essential for recovering segments that capture persistent behavioral differences.

5.4.2 Accounting for Correlated Preferences within Users

While the posterior network infers segments from individual trajectories, unconstrained training implicitly treats each trajectory as independent, ignoring correlation across those generated by the same user. This neglects a central economic feature of consumption behavior: repeated decisions by the same individual are likely governed by a common latent preference. Such independence is at odds with models of persistent heterogeneity (Arcidiacono and Jones, 2003) and with empirical evidence that preferences tend to be stable across products and contexts (Ainslie and Rossi, 1998; Seetharaman et al., 1999).[7]

A natural solution—pooling posterior predictions across a user's trajectories, as in EM-style latent class models (Arcidiacono and Jones, 2003)—proved unstable in our setting, collapsing to a single segment. To address this, we introduce a contrastive regularization objective inspired by InfoNCE (Oord et al., 2018). The key idea is to encourage posterior segment distributions to be similar across trajectories from the same user, while *discouraging such similarity across trajectories from different users.* This contrastive formulation enforces within-user consistency while preserving separation across users.

Formally, let $w_i = q_\psi(z \mid \tau_i)$ denote the posterior distribution for trajectory τ_i, and let w_j correspond to another trajectory from the same user. For each anchor w_i, we maximize similarity with w_j relative to trajectories from other users:

$$\mathcal{L}_{\text{CR}} = \mathbb{E}_{\mathcal{U}} \left[-\log \frac{\exp\big(\text{sim}(w_i, w_j)/r\big)}{\sum_{k \neq i} \exp\big(\text{sim}(w_i, w_k)/r\big)} \right], \tag{7}$$

where $\text{sim}(\cdot)$ denotes cosine similarity, r is a temperature parameter, and the expectation is taken over users in the observed data.[8]

[7] Much of multi-category choice literature focuses on static rather than forward-looking dynamic choice.

[8] From an information-theoretic perspective, this objective maximizes mutual information between user identity and segment assignment; see Oord et al. (2018). We set $r = 0.5$ following standard practice.

The posterior network is trained by minimizing the combined objective

$$\mathcal{L}_{\text{posterior}} = \mathcal{L}_{\text{CE}} + \lambda_{CR} \cdot \mathcal{L}_{\text{CR}}. \tag{8}$$

where λ_{CR} controls the strength of the consistency regularization. This formulation allows us to recover stable, interpretable latent segments without relying on joint likelihood estimation or explicit user-level latent variables.

5.5 S4. Reward Assignment

Following discriminator and posterior updates, we assign scalar rewards to each transition in simulated trajectories. The total reward combines an adversarial component from the discriminator and a mutual information component from the posterior network.

For a transition (s, a, s') in trajectory τ with latent segment z, the shaped reward is

$$\tilde{r}(s, a, s', z) = r_{\text{adv}}(s, a, s', z) + \lambda_{MI} \cdot \overline{r}_{MI}(\tau),$$

$$\text{where} \quad r_{\text{adv}}(s, a, s', z) = f_{\theta,\phi}(s, a, s', z) - \log \pi(a \mid s, z) \quad \text{and} \quad r_{MI}(\tau) = \log q_{\psi}(z \mid \tau).$$

$\overline{r}_{MI}(\tau)$ denotes the standardized version of $r_{MI}(\tau)$ and λ_{MI} controls the relative importance. The adversarial reward encourages the policy to generate transitions that are consistent with observed behavior under the current reward estimate, while the mutual information reward incentivizes trajectories that are informative about the latent segment.

Substituting $\tilde{r}$ into the policy optimization problem yields

$$\max_{\pi} \; \mathbb{E}_{\pi} \left[\sum_{t=0}^{T} f_{\theta,\phi}(s_t, a_t, s'_t, z) + \lambda_{MI} \cdot \overline{r}_{MI}(\tau) \right] + \mathcal{H}(\pi).$$

In practice, posterior segment assignments stabilize substantially earlier than the discriminator and policy. Once convergence is reached, we freeze the posterior network and set $\lambda_{MI} = 0$, reducing the objective to the standard entropy-regularized AIRL objective.

5.6 S5. Policy Update via Forward Reinforcement Learning

The final step updates the policy to align with the segment-specific rewards recovered in Step S4. For each segment z, the policy $\pi(a \mid s, z)$ is updated to solve a forward RL problem,

$$\max_{\pi} \mathbb{E}_{\pi} \left[\sum_{t=0}^{T} \tilde{r}_t(s_t, a_t, s'_t, z) \right].$$

We use proximal policy optimization (PPO; Schulman et al. 2017) for its computational efficiency and stability, though any gradient-based policy optimization method can be used. PPO updates the policy using simulated trajectories while constraining successive policy changes to ensure stable learning. We follow standard PPO procedures outlined in Online Appendix Section W6. This ensures consistency between the reward recovered through adversarial training and the behavior induced by the learned policy.

At convergence, the learned policy represents the optimal response to the recovered segment-specific reward function. Full algorithmic details, including network architectures, optimization schedules, and hyperparameters, are provided in Online Appendix Section W7.

6 Empirical Results

This section presents the empirical results of the estimated model. We first examine the recovered latent segments, highlighting systematic differences in consumption behavior and sensitivity to key state variables. We then assess external validity using outcomes from a separate field experiment. Additional model performance comparisons are reported in Online Appendix Section W8.

6.1 Interpreting Latent Segments

We begin by examining aggregate consumption patterns across latent segments. The number of segments is fixed at $K = 4$, balancing flexibility and parsimony. Figure 4 summarizes key

engagement metrics—including series completion rates, purchase probabilities, and total episodes consumed and purchased—revealing sharp differences in consumption styles.

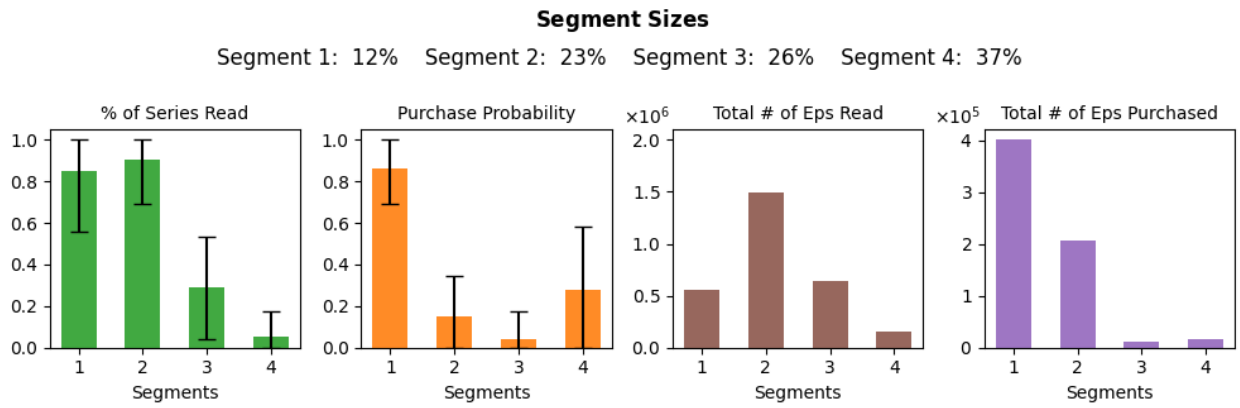

Figure 4: Aggregate consumption statistics across four latent segments.

Segment 1 (12% of users) is the most engaged and economically valuable: despite its small size, it accounts for 63% of all purchases and exhibits the highest completion and purchase rates. Segment 2 (23% of users) also achieves high completion rates but differs markedly in its spending behavior. Although these users purchase episodes to a much lesser extent, they remain highly engaged, generating 52% of total consumption. In contrast, Segments 3 and 4 contribute minimally to engagement and revenue, typically exiting after a few free episodes. Together, Segments 1 and 2 account for 72% of total consumption and 96% of purchases. Accordingly, we focus subsequent analyses on these two high-engagement groups, referred to as the *Pay & Read* and *Wait & Read* segments.

Having established descriptive differences, we next examine how segments respond to changes in state variables. We estimate policy sensitivities using a perturbation-based approach that computes local marginal effects of a one-standard-deviation increase in each structured state variable on action probabilities, holding other inputs fixed (Puri et al., 2019; Ivanovs et al., 2021).[9] Effects are averaged across representative states, with standard errors obtained via bootstrap. This method provides an interpretable summary of how each

[9]For binary variables such as `Price` and `Prev Pay`, the reported effect corresponds to the discrete change from 0 to 1. Restricting the analysis to WFF episodes allows these sensitivities to be interpreted directly as changes in the likelihood of purchase, wait, or exit.

segment trades off price, wait time, and content features.

Table 4 reports the resulting sensitivities for the *Pay & Read* and *Wait & Read* segments. When an episode shifts from free to WFF pricing, both segments become less likely to consume immediately, consistent with a negative price coefficient. However, the magnitude differs sharply: the *Pay & Read* segment is 12% less likely to purchase, whereas the *Wait & Read* segment experiences a 77% drop—indicating much stronger aversion to paying.

Both segments respond to longer wait-times by shifting away from delayed access, but the adjustment occurs along different margins. For the *Pay & Read* segment, longer delays primarily induce substitution from waiting to purchasing, with only a small increase in exit. In contrast, while the *Wait & Read* segment also becomes more likely to purchase, the increase in exit is more pronounced, reflecting their stronger reluctance to pay and greater sensitivity to prolonged delays as a deterrent to continued engagement.

Narrative features affect action choices in systematically different ways across segments. For the *Pay & Read* segment, increases in narrative intensity—such as stronger cliffhangers and faster narrative speed—shift behavior away from waiting/exit and toward purchase. In contrast, increases in narrative volume and circuitousness instead encourage waiting, consistent with these readers' greater tolerance for deferral when content density is high. Emotional tone has a more muted or offsetting effect.

In contrast, the *Wait & Read* segment exhibits a markedly muted response to narrative features. Coefficient magnitudes are substantially smaller across all content dimensions. This pattern suggests that consumption decisions for these readers are driven less by content variation and more by structured constraints such as price and wait-time, consistent with a segment whose behavior is governed by access frictions rather than narrative appeal.

6.2 Validation on Field Experiment Data

As described in Section 3.2, the platform implemented an exogenous reduction in wait times for a subset of series. This intervention provides a rare opportunity to evaluate the model's

	Pay & Read Segment			Wait & Read Segment		
	Purchase	*Wait*	*Exit*	*Purchase*	*Wait*	*Exit*
Price	-0.124***	0.123***	0.001	-0.773***	0.772***	0.001
	(0.005)	(0.005)	(0.001)	(0.004)	(0.005)	(0.000)
Wait-time	0.140***	-0.149***	0.009***	0.083***	-0.105***	0.022***
	(0.003)	(0.003)	(0.001)	(0.002)	(0.002)	(0.001)
Progress	-0.004***	0.005***	-0.001*	-0.005***	0.008***	-0.003***
	(0.001)	(0.001)	(0.000)	(0.000)	(0.000)	(0.000)
Prev Pay	0.353***	-0.332***	-0.022***	0.482***	-0.484***	0.003***
	(0.004)	(0.004)	(0.001)	(0.003)	(0.003)	(0.000)
Cliffhanger	0.013***	-0.012***	-0.001***	0.006***	-0.006***	-0.000
	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)
Narr. Speed	0.038***	-0.033***	-0.005***	-0.001***	0.003***	-0.002***
	(0.001)	(0.001)	(0.000)	(0.000)	(0.000)	(0.000)
Narr. Volume	-0.043***	0.038***	0.005***	-0.002***	-0.000	0.002***
	(0.001)	(0.001)	(0.000)	(0.000)	(0.000)	(0.000)
Narr. Circ.	-0.016***	0.017***	-0.001***	-0.000*	0.000	0.000***
	(0.001)	(0.001)	(0.000)	(0.000)	(0.000)	(0.000)
Valence	-0.007***	0.006***	0.001***	-0.003***	0.002***	0.001***
	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)
Arousal	-0.004***	0.002***	0.002***	0.001***	-0.001***	-0.000***
	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)

$^{*}p < 0.05$; $^{**}p < 0.01$; $^{***}p < 0.001$.

Table 4: Sensitivity of $\pi(a \mid s, z)$ to perturbations in structured state features. Reported values represent the average marginal effect of a one-standard-deviation increase in each feature on the probability of immediate consumption, holding all other inputs fixed. Standard errors are computed via 100 nonparametric bootstrap repetitions.

out-of-sample predictive validity under real-world policy variation. A key strength of our framework is that it accommodates latent segment heterogeneity, allowing predictions to be evaluated not only in aggregate but also at the segment level. All affected series are excluded from model estimation, allowing us to assess whether the model generalizes to unseen content and accurately predicts behavioral responses to changes in access frictions. Such external validations remain uncommon in the DDC literature, with notable exceptions including Todd and Wolpin (2006) and Misra and Nair (2011).

We conduct the evaluation by simulating consumption trajectories under both the original and reduced wait times, matching the number of users per series and per latent segment observed in the data. Segment membership is inferred from users' training-set trajectories, and predicted outcomes are aggregated to the segment and series levels. We then compare

predicted and realized changes in consumption before and after the policy intervention.

The left panel of Figure 5 reports the model-predicted changes. For both segments, shorter wait times increase completion rates while reducing purchase probabilities, reflecting substitution toward free consumption when access frictions are relaxed. Although purchase levels remain higher for the *Pay & Read* segment, the proportional decline in purchasing is similar across segments. These predictions align closely with the estimated policy sensitivities in Table 4, which indicate that longer waits increase purchase incentives, while shorter delays promote continued consumption.

The right panel of Figure 5 shows the corresponding changes observed in the field experiment. The close correspondence between predicted and realized responses—both in direction and magnitude—provides strong evidence that the model captures how users adjust behavior in response to changes in wait-time.

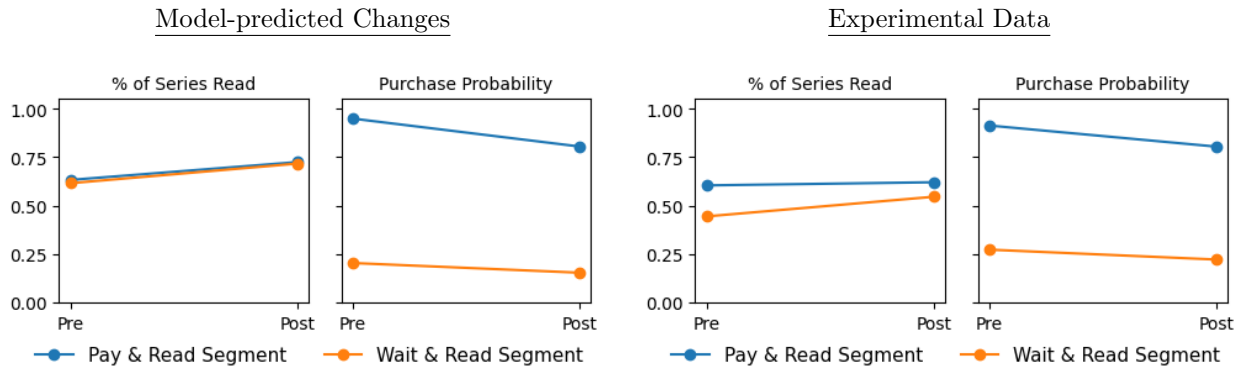

Figure 5: **Left:** Model-predicted change in consumption before vs. after wait-time reduction. **Right:** Actual change in consumption from field experiment data.

Notably, the model not only captures the direction and magnitude of these behavioral changes but also closely aligns with the absolute levels of consumption before and after the intervention, as reported in Table 5. These findings provide strong validation of the structural model, lending credibility to its use in counterfactual analyses.

	(Aggregate) MAPE		(Series-level) MAE	
	# of Eps Read	# of Eps Purchased	Purchase Prob	% of Series Read
Pre-reduction	0.18	0.18	0.19	0.16
Post-reduction	0.09	0.13	0.13	0.10

Table 5: Model evaluation metrics before and after wait-time reduction

7 Counterfactual Analysis

We use the estimated model to evaluate user consumption and purchase behavior under counterfactual platform access rules. Specifically, we re-optimize the decision policy using the learned reward function within the modified MDP environment, when these changes alter users' expectations about future access or prices, and then simulate user trajectories under the resulting conditions. For each scenario, we generate the same number of trajectories per segment and per series from the training set to ensure comparability across conditions.

7.1 Can Free Increase Pay?

We first examine how the platform's WFF strategy shapes both consumption and purchases. Standard versioning and freemium logic predicts that WFF should expand total consumption but reduce monetization, as many users substitute from paid to free access. We test this prediction by simulating a counterfactual environment without WFF, where all delayed-free episodes are converted to paid-only while initial free episodes remain unchanged.

Figure 6 shows that this prediction holds for the *Wait & Read* segment: WFF increases per-user consumption by 121% but reduces purchases by 56%. Strikingly, however, the *Pay & Read* segment experiences gains on both fronts—consumption rises by 52% and purchases by 36%. This runs counter to the classic versioning prediction that free access uniformly cannibalizes paid demand.

Aggregating across segments, WFF increases total consumption by 96% and reduces purchases by 18%. These represent a conservative estimate because our model abstracts from the decision to start a series. The counterfactual holds the size of the reader base

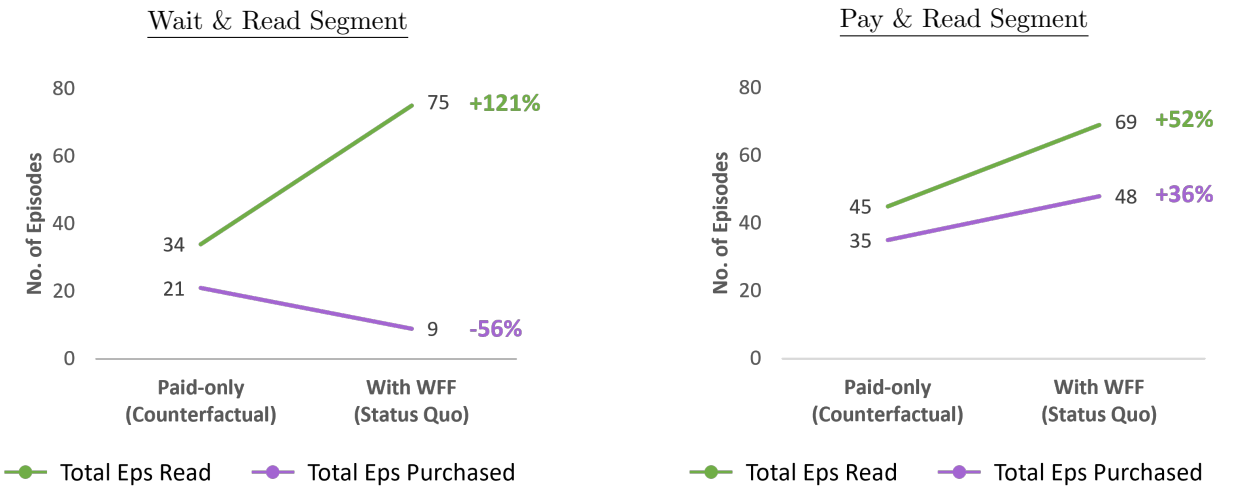

Figure 6: Per user consumption statistics without (counterfactual) and with (status quo) WFF policy by segment

fixed, capturing only within-user changes. Prior work shows that WFF also expands entry, implying larger total effects on engagement and revenue (Choi et al., 2025; Lee et al., 2025).

WFF effects vary substantially across series (Figure 7): some series experience substantial gains in both consumption and purchases, while others see muted or even negative responses. Motivated by Lee et al. (2025), we test whether this heterogeneity is linked to narrative dynamics: if WFF helps users move past weaker stretches of a series, then the largest WFF-driven gains should occur in series with lower cliffhanger strength.[10] We evaluate this by comparing cliffhanger scores across series in the top versus bottom tercile (top 33% vs. bottom 33%) of predicted WFF gains. Consistent with the mechanism, high-gain series have significantly lower cliffhanger strength than low-gain series both for consumption gains ($t = 4.37$, $p < 0.001$) and purchase gains ($t = 7.03$, $p < 0.001$).

Together, these findings highlight the central role of content in shaping monetization outcomes. The effect of free access is not uniform: while it can cannibalize spending for some users, for others it does the opposite, helping them move past weaker episodes and ultimately generating more paid engagement. Thus, WFF is not merely a tool to sustain engagement, but a mechanism that can actively translate free consumption into higher purchases.

[10]See Section 5.1 and Online Appendix Section W5 for details on the cliffhanger score.

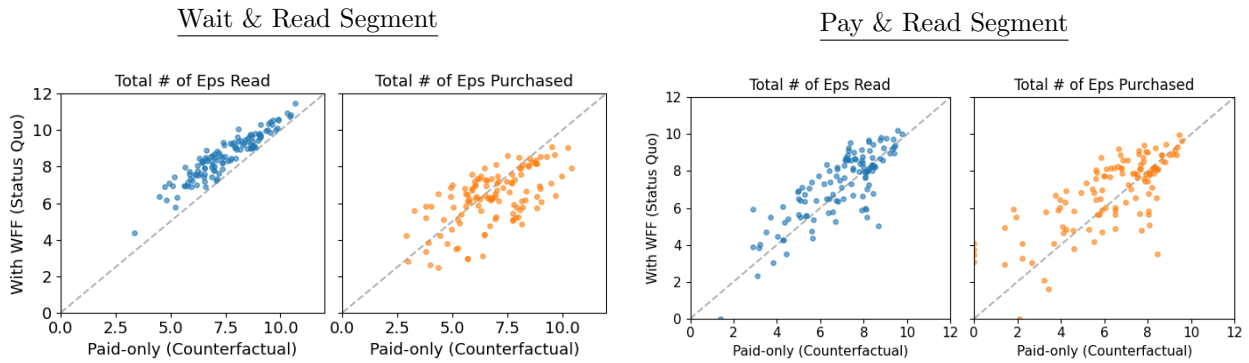

Figure 7: Series-level consumption and purchases by segment without and with WFF policy. Each dot represents a series. Dashed line indicates no change (i.e., $y = x$).

7.2 Can Patience be a Discrimination Lever?

The first counterfactual showed that WFF has two faces: it can cannibalize purchases for some users but expand monetization for others. This raises a natural question: how can platforms reduce cannibalization while preserving the monetization benefits of delayed free access? We address this by exploiting heterogeneity in users' *willingness to wait*. Using the segment-level sensitivities to price and delay documented in Section 6.1, we evaluate how access frictions can be used to screen users into exit versus paid conversion.

Although directly implementing personalized wait-times may be infeasible due to fairness concerns or potential consumer backlash, the simulation offers practical insights for targeted strategies. For instance, platforms may design promotional campaigns that reduce wait-times selectively for disengaging users, or provide fast-track access as a loyalty incentive.

To evaluate the role of patience as a screening device, we simulate user trajectories under alternative MDP environments with wait times of 1, 12, and 24 hours, holding all other features fixed. Figure 8 summarizes consumption and purchase outcomes across segments.

For the *Wait & Read* segment, longer wait-times sharply reduce engagement: total episodes read falls monotonically as delays increase. At the same time, purchases become more likely conditional on continuing—purchase probability rises with wait-time, and the total number of purchased episodes increases. This pattern mirrors the local perturbation estimates: longer delays shift behavior away from waiting and toward both purchasing and

exit. In other words, delay acts as a screening device within this segment, inducing early churn among low-patience users while increasing conversion among those who remain active.

For the *Pay & Read* segment, longer wait-times also increase purchase intensity, but overall consumption declines. Both total episodes read and the number of purchased episodes fall as delays increase, indicating that the dominant effect is reduced reading volume, even though remaining consumption becomes more purchase-intensive.

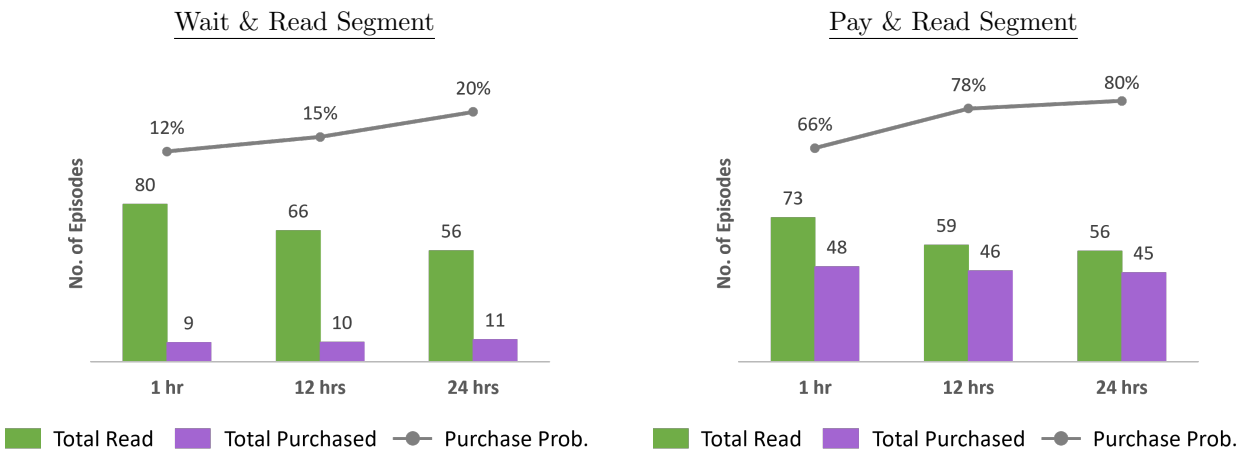

Figure 8: Per-user consumption statistics by wait-time by segment

Guided by these patterns, we evaluate a segment-specific access policy designed to maximize monetization. Under this policy, *Wait & Read* users face a 24-hour delay, while *Pay & Read* users face a 1-hour delay. Relative to the status quo, this design increases total purchases by 6.9%, despite a 16.6% decline in total consumption. Taken together, the results illustrate how wait-time can act as a screening device: it disproportionately pushes low-patience users to exit, while shifting remaining consumption toward paid access. Segment-aligned delays therefore increase monetization at the cost of lower overall engagement.

7.3 Can Content Guide Pricing?

The first two counterfactuals show that WFF's effects depend on *who* the user is: for some it cannibalizes, for others it expands monetization, and segment-based delays can shift that balance. But heterogeneity also arises *within* a series: some episodes are low-stakes fillers,

while others create peak engagement. We therefore evaluate a content-based pricing policy that keeps weaker episodes under WFF to sustain progression while setting high-engagement episodes as paid-only to monetize peak willingness to pay.

We operationalize this using the learned policy. To isolate the role of content, we remove price variation by setting all episodes to WFF and compute segment-specific action probabilities $\pi(a \mid s, z)$. For each episode, we define its *purchase margin*:

$$\Delta(s, z) = \pi(a = 0 \mid s, z) - \max\{\pi(a = 1 \mid s, z), \pi(a = 2 \mid s, z)\},$$

where $a = 0$ denotes purchase, $a = 1$ waiting, and $a = 2$ exit. Episodes with higher $\Delta(s, z)$ are those where users are most inclined to purchase rather than wait or exit.

We sort episodes by $\Delta(s, z)$ and consider quantile-based thresholds over its distribution. For a given cutoff q, episodes above the qth percentile are designated paid-only, while those below remain WFF.[11] Lower cutoffs monetize more episodes; higher cutoffs concentrate paywalls on only the strongest purchase-margin moments.

Figure 9 reports simulated outcomes across thresholds. For the *Wait & Read* segment, consumption increases as more episodes remain WFF, reflecting reliance on delayed free access. Purchases, however, follow a non-monotonic path: they rise up to the 30th quantile, where weaker episodes are available under WFF while high-engagement episodes remain paid-only, and then decline as WFF access expands. At the 30th quantile, per-user consumption remains essentially unchanged relative to the status quo, while purchases increase by 225%.

For the *Pay & Read* segment, expanding WFF availability increases both consumption and purchasing relative to the status quo. By reducing access frictions for more episodes, users reach later parts of the series more often, which generates additional paid interactions. The relationship is not strictly monotone, but improvements are stable across a wide range of thresholds. At the peak cutoff of 50th quantile, consumption and purchases rise by 20% and 18%, respectively. Compared with the *Wait & Read* segment, responses are more muted because these users already exhibit high willingness to pay, so the policy shifts behavior on

[11] Under the status quo, 82% of episodes are WFF and 18% paid-only.

the margin rather than producing large substitution effects.

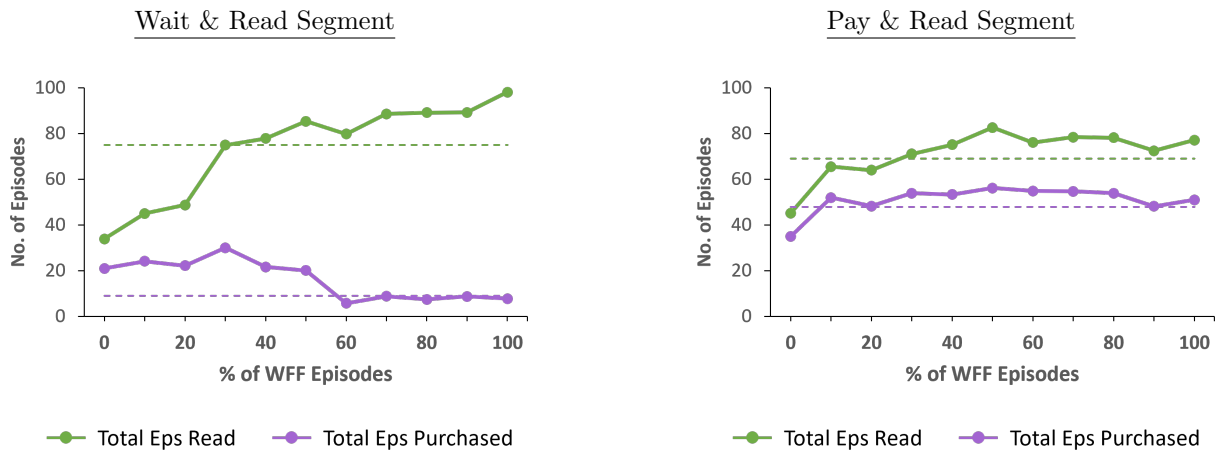

Figure 9: Per-user consumption statistics under content-based pricing policies by segment. x-axis indicates share of episodes designated as WFF. Dashed lines represent status quo.

Overall, the counterfactual highlights the value of content-aware monetization: paywalls are most effective when placed at episodes where users are intrinsically willing to pay, while free access to weaker episodes helps preserve progression and unlock downstream conversion opportunities. This approach offers a middle ground between broad free access and strict paywalls, maximizing both engagement and revenue potential.

8 Conclusion

This paper develops a novel framework for modeling consumption in high-dimensional content environments and applies it to evaluate monetization strategies for serialized media. Our counterfactual analyses reveal interesting insights. First, wait-for-free access, initially expected to cannibalize purchases within the user, can paradoxically expand monetization by helping high-spending users progress through weaker episodes and reach later points of paid engagement. Second, customizing wait-times by segment reduces cannibalization and substantially amplifies purchases, showing how patience can be leveraged to rebalance retention and revenue. Third, content itself can guide pricing: paywalls placed at high-engagement episodes maximize monetization while leaving weaker episodes free sustains progression.

Methodologically, we extend AIRL to incorporate latent user heterogeneity and introduce a corpus-scale content representation that improves generalization across diverse narratives. While our application focuses on serialized text, the framework extends to other domains where consumption unfolds sequentially, such as television shows, podcasts, or video games. More broadly, the idea of adversarial estimation is not limited to dynamic choice models with unstructured content. It can also be applied to static models or to settings with only structured variables but high-dimensional state spaces, providing a flexible approach to recover structural primitives when conventional likelihood-based methods become intractable.

Our framework equips platforms with a powerful tool to evaluate content and monetization strategies. It enables assessment of how users will respond to new content, pricing strategies, and targeted promotions—accounting for both user heterogeneity and the evolving narrative content. It also supports recommendation systems that move beyond static preference matching to reflect interdependencies within serialized content.

Despite these contributions, our work has several limitations. First, the model focuses on within-series engagement and does not capture the user's initial decision to start a series—a decision that depends on the choice set, search behavior, and platform recommendations. That said, our framework is well-suited to serve as the second stage of a two-part model, with the first stage capturing series selection. Second, the model treats each user–series trajectory as independent and does not account for cross-series substitution or opportunity cost. Addressing this would require a joint state representation that captures parallel content consumption or a coupled system of trajectories in which rewards in one series depend on states in others—both of which introduce substantial computational complexity and fall outside the scope of our framework. Third, the model abstracts from social influences such as peer effects or word-of-mouth, which may shape consumption decisions. Finally, our analysis is limited to binary variation in price and cannot address continuous price sensitivities.

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Web Appendix

W1 Inverse Reinforcement Learning Background

In standard reinforcement learning (RL), the goal is to learn a policy π^* that maximizes the expected sum of discounted rewards, assuming the reward function is known. Prior work in marketing has used RL to study optimal policies when rewards are directly observed—such as game points, advertising clicks, or consumer spending (Misra et al., 2019; Liu, 2023; Wang et al., 2023; Ma et al., 2025). However, in many settings, the true reward function that governs behavior is latent and not directly measurable.

Inverse reinforcement learning (IRL) (Ng and Russell, 2000; Abbeel and Ng, 2004) instead seeks to recover the reward function $r^*(s, a)$ that best explains observed decision sequences $\mathcal{D} = \{\tau_1, ..., \tau_N\}$. Once the reward is learned, the optimal policy π^* can be recovered through RL. Maximum Entropy (MaxEnt) IRL (Ziebart et al., 2008), the most widely adopted framework, assumes that among all policies consistent with observed behavior, the one that is most uncertain—or has maximum entropy—is preferred:

$$\max_{\theta} \left[\mathbb{E}_{(s,a)\sim\mathcal{D}} \left[r_\theta(s, a)\right] - \mathrm{RL}(r_\theta)\right] \tag{9}$$

$$\text{where} \quad \mathrm{RL}(r_\theta) = \max_{\pi} \left[\mathbb{E}_{(s,a)\sim\pi} \left[r_\theta(s, a)\right] + \mathcal{H}(\pi)\right] \tag{10}$$

Intuitively, this formulation searches for a reward function under which the observed behavior yields the highest expected rewards, while ensuring that no other policy can do better once both rewards and entropy are taken into account. This prevents the model from simply assigning overly sharp rewards that explain the given data but do not generalize.

Another way to interpret this framework is to view agents as making stochastic choices, where the probability of a trajectory increases with its total expected reward. Specifically, we assume that the probability of observing a trajectory τ is proportional to the exponential of its cumulative reward. This allows inverse reinforcement learning to be cast as a maximum

likelihood estimation (MLE) problem:

$$\max_{\theta} \mathbb{E}_{\tau \sim \mathcal{D}}[\log p_\theta(\tau)] \quad \text{where} \quad p_\theta(\tau) = \frac{1}{Z_\theta} \exp\Big(r_\theta(\tau)\Big), \quad Z_\theta = \int \exp\Big(r_\theta(\tau)\Big) d\tau.$$

Here, $p_\theta(\tau)$ is the probability of trajectory τ under the model-implied reward function r_θ, and Z_θ is the normalizing constant (also known as the partition function) that ensures $p_\theta(\tau)$ defines a valid probability distribution.

Assuming a Markov decision process, the optimal policy under this framework takes the form of a softmax over state-action values:

$$p_\theta(a \mid s) = \frac{\exp\left(Q_\theta(s,a)\right)}{\sum_{a'} \exp\left(Q_\theta(s,a')\right)},$$

where $Q_\theta(s,a) = r_\theta(s,a) + \gamma \mathbb{E}_{s'}[V_\theta(s')]$ and $V_\theta(s) = \log \sum_{a'} \exp\left(Q_\theta(s,a')\right)$. This value function $V_\theta(s)$ corresponds to the inclusive value under T1EV errors.

Although developed independently, MaxEnt IRL is mathematically equivalent to maximum likelihood estimation in DDC models, when the choice model follows a logit specification (Rust, 1987; Hotz and Miller, 1993; Aguirregabiria and Mira, 2002). In both frameworks, the objective is to find reward or utility parameters that maximize the likelihood of observed choices. However, as with DDC models, MaxEnt IRL faces scalability challenges in large state spaces due to the need to explicitly evaluate the likelihood.[12]

[12]Specifically, computing Z_θ involves integrating over all possible trajectories, which becomes intractable in large or continuous state spaces.

W2 Mobile Application Interface

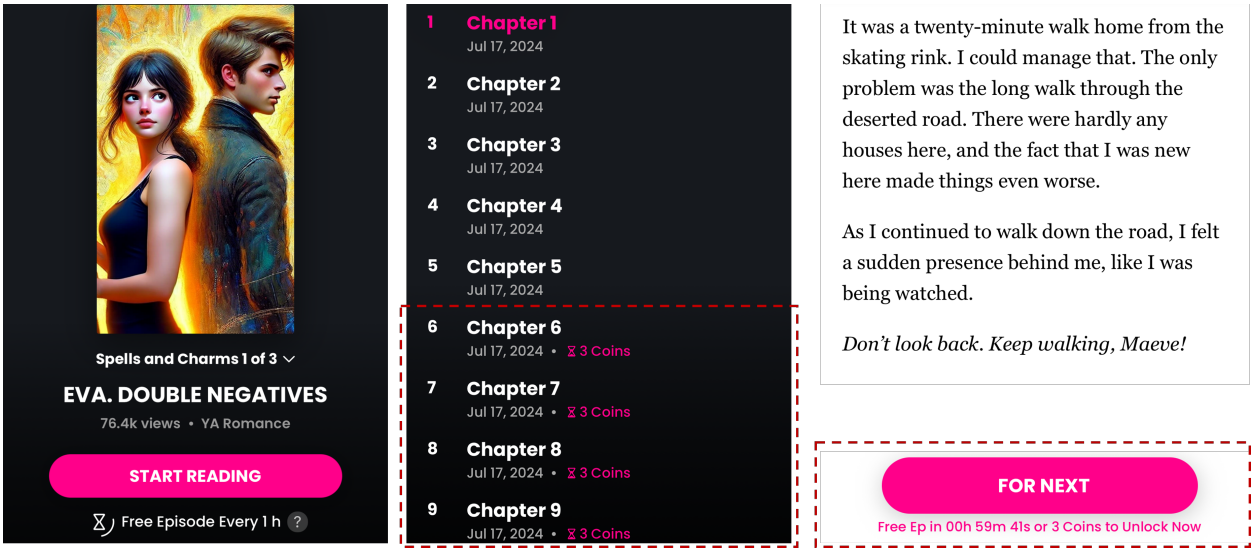

Figure W2.1: User interface of the serialized novel platform

W3 Identification of Action-dependent Rewards

Setup. We consider the following environment:

- Finite state space $\mathcal{S}$, and action set $\mathcal{A}$ that includes an exit action at every state.
- Exit deterministically transitions to absorbing state s_{abs}: $P(s_{\text{abs}} \mid s, \text{exit}) = 1 \quad \forall s \in \mathcal{S}$.
- Known anchors: $g(s, \text{exit}) = 0 \quad \forall s \in \mathcal{S}, \quad h(s_{\text{abs}}) = 0$.
- Discount factor $\beta \in (0, 1)$ is known.
- AIRL discriminator is parameterized as: $f(s, a, s') = g(s, a) + \beta h(s') - h(s)$.
- At optimum, the learned f equals the advantage: $A^*(s, a) = r^*(s, a) + \beta V^*(s') - V^*(s)$.

We prove two results:

(i) **Uniqueness of the (g, h)-decomposition:** Given a fixed f, the anchors force a unique pair (g, h) that realizes f.

(ii) **Recovery of the true utilities:** If the true per-period utility is $r^*(s, a)$ and the true value is $V^*(s)$ with the same exit/absorbing levels ($r^*(s, \text{exit}) = 0$, $V^*(s_{\text{abs}}) = 0$), then the unique AIRL pair equals the truth: $g = r^*$ and $h = V^*$.

Lemma 1 (General Non-identifiability)**.** *Suppose two pairs (g, h) and $(\tilde{g}, \tilde{h})$ produce the same AIRL score f for all triples (s, a, s'):*

$$g(s, a) + \beta h(s') - h(s) = \tilde{g}(s, a) + \beta \tilde{h}(s') - \tilde{h}(s).$$

Define the state function $\Phi(s) := \tilde{h}(s) - h(s)$. Then:

$$\tilde{h}(s) = h(s) + \Phi(s), \qquad \tilde{g}(s, a) = g(s, a) + \Phi(s) - \beta \Phi(s').$$

Conversely, for any Φ, setting $\tilde{h}, \tilde{g}$ in this way yields identical AIRL scores.

Proof. Start from equality of scores:

$$g(s,a) + \beta h(s') - h(s) = \tilde{g}(s,a) + \beta \tilde{h}(s') - \tilde{h}(s).$$

Rearrange to isolate $\tilde{g} - g$:

$$\tilde{g}(s,a) - g(s,a) = \beta\big(h(s') - \tilde{h}(s')\big) - \big(h(s) - \tilde{h}(s)\big).$$

By definition, $\Phi(s) = \tilde{h}(s) - h(s)$, so $h(s) - \tilde{h}(s) = -\Phi(s)$. Substituting gives:

$$\tilde{g}(s,a) = g(s,a) + \Phi(s) - \beta\Phi(s').$$

Finally,

$$\begin{aligned}\tilde{g}(s,a) + \beta\tilde{h}(s') - \tilde{h}(s) &= \big[g(s,a) + \Phi(s) - \beta\Phi(s')\big] + \beta\big[h(s') + \Phi(s')\big] - \big[h(s) + \Phi(s)\big] \\ &= g(s,a) + \beta h(s') - h(s).\end{aligned}$$

Hence the two scores match for all (s, a, s'). In other words, (g, h) is not uniquely identified.

□

Theorem 1 (Identification with known anchors)**.** *Assume the anchors $g(s, exit) \equiv 0$ for all $s \in \mathcal{S}$ and $h(s_{abs}) \equiv 0$. If (g, h) and $(\tilde{g}, \tilde{h})$ both realize the same f, then $g = \tilde{g}$ and $h = \tilde{h}$.*

Proof. By Lemma 1, there exists a function Φ such that

$$\tilde{h}(s) = h(s) + \Phi(s), \qquad \tilde{g}(s,a) = g(s,a) + \Phi(s) - \beta\Phi(s').$$

Apply the exit anchor at each s:

$$0 = \tilde{g}(s, \text{exit}) = g(s, \text{exit}) + \Phi(s) - \beta\Phi(s_{\text{abs}}) = 0 + \Phi(s) - \beta\Phi(s_{\text{abs}}).$$

Hence, $\Phi(s) = \beta\Phi(s_{\text{abs}})$ for all s. Apply the absorbing state anchor:

$$0 = \tilde{h}(s_{abs}) = h(s_{abs}) + \Phi(s_{abs}) = 0 + \Phi(s_{abs}),$$

so $\Phi(s_{abs}) = 0$, and hence $\Phi(s) = 0$ for all s. Therefore, $\tilde{h} = h$ and $\tilde{g} = g$. □

Theorem 2 (Identification of the underlying utilities)**.** *Suppose the true Bellman pair* (r^*, V^*) *satisfies, for all* (s, a, s')*,*

$$A^*(s, a) = r^*(s, a) + \beta V^*(s') - V^*(s),$$

with $r^*(s, exit) = 0$ *and* $V^*(s_{abs}) = 0$*.*

Let (g, h) *be any AIRL pair realizing the same advantage, i.e.,*

$$A^*(s, a) = g(s, a) + \beta h(s') - h(s),$$

and satisfying the same anchors $g(s, exit) = 0, h(s_{abs}) = 0$*. Then* $g = r^*$ *and* $h = V^*$*.*

Proof. Subtract the two representations $A^*(s, a)$:

$$(g - r^*)(s, a) + \beta(h - V^*)(s') - (h - V^*)(s) = 0 \quad \forall (s, a, s').$$

Define $\Phi \coloneqq h - V^*$. Then

$$(g - r^*)(s, a) = \Phi(s) - \beta\Phi(s').$$

By the exit and absorbing state anchors (applied to both pairs),

$$0 = (g - r^*)(s, a) = \beta\Phi(s_{abs}) - \Phi(s) \quad \text{and} \quad \Phi(s_{abs}) = h(s_{abs}) - V^*(s_{abs}) = 0$$

so $\Phi \equiv 0$, and hence $h = V^*$ and $g = r^*$. □

W4 Extension of Section 1 for Stochastic Transitions

The goal is to prove that $g = r^*$ and $h = V^*$ under stochastic settings given the anchors. Let the agent's true optimization problem be a (soft) MDP defined by:

- **Advantage:** $A^*(s,a) = r^*(s,a) + \beta \mathbb{E}_{T(s'|s,a)}[V^*(s')] - V^*(s)$
- **Transition Kernel:** $T(s' \mid s,a)$, the agent's belief over the next state.

Let the AIRL discriminator, f, be parameterized by learned functions g and h:

- **AIRL Discriminator:** $f(s,a,s') = g(s,a) + \beta h(s') - h(s)$

At optimality, the *expected value* of the discriminator (taken over the transitions T) matches the true Advantage:

- **Optimality Condition:** $\mathbb{E}_{T(s'|s,a)}[f(s,a,s')] = A^*(s,a)$

As before, the same anchors apply:

- Exit deterministically transitions to absorbing state s_{abs}: $P(s_{\text{abs}} \mid s, \text{exit}) = 1 \quad \forall s \in \mathcal{S}$.
- Known anchors: $g(s, \text{exit}) = 0 \quad \forall s \in \mathcal{S}, \quad h(s_{\text{abs}}) = 0$.

Theorem 3. *Given these definitions and assumptions, the AIRL optimum uniquely identifies the true reward and value functions: $g = r^*$ and $h = V^*$.*

Proof. Start with the optimality condition, $\mathbb{E}_T[f(s,a,s')] = A^*(s,a)$, and substitute the definitions of f and A^*:

$$g(s,a) + \beta\, \mathbb{E}_T[h(s') \mid s,a] - h(s) = r^*(s,a) + \beta\, \mathbb{E}_T[V^*(s') \mid s,a] - V^*(s)$$

Rearrange the equation to isolate terms by g/r^* and h/V^*:

$$(g(s,a) - r^*(s,a)) = (h(s) - V^*(s)) - \beta\, (\mathbb{E}_T[h(s') \mid s,a] - \mathbb{E}_T[V^*(s') \mid s,a])$$

Define $\Phi(s) := h(s) - V^*(s)$, and substitute $\Phi(s)$ into this equation.

$$(g(s,a) - r^*(s,a)) = \Phi(s) - \beta\, \mathbb{E}_T[\Phi(s') \mid s, a] \tag{11}$$

We evaluate the equation for $a = \text{exit}$:

$$(g(s,\text{exit}) - r^*(s,\text{exit})) = \Phi(s) - \beta\, \mathbb{E}_T[\Phi(s') \mid s, \text{exit}]$$

Applying the exit reward anchor, the left side of the equation becomes $(0 - 0) = 0$.

$$0 = \Phi(s) - \beta\, \mathbb{E}_T[\Phi(s') \mid s, \text{exit}]$$

Because the exit transition is deterministic, the expectation $\mathbb{E}_T[\Phi(s') \mid s, \text{exit}]$ collapses to a single, known outcome: $\Phi(s_{abs})$. Hence,

$$\Phi(s) = \beta\, \Phi(s_{abs}) \tag{12}$$

Applying the absorbing state anchor, we find the value of $\Phi(s_{abs})$ from its definition:

$$\Phi(s_{abs}) = h(s_{abs}) - V^*(s_{abs}) = 0 - 0 = 0$$

Substituting $\Phi(s_{abs}) = 0$ into Equation 12 gives:

$$\Phi(s) = \beta \cdot 0 \quad \Rightarrow \quad \Phi(s) = 0 \quad \forall s.$$

Since $\Phi(s) = h(s) - V^*(s)$, this gives $h(s) = V^*(s)$. Substitute $\Phi(s) = 0$ into Equation 11:

$$(g(s,a) - r^*(s,a)) = (0) - \beta\, \mathbb{E}_T[0 \mid s, a] \quad \Rightarrow \quad (g(s,a) - r^*(s,a)) = 0$$

This gives $g(s,a) = r^*(s,a)$. $\square$

W5 Structured Content Features

The strength of the cliffhanger is measured using `Llama-4-Maverick-17B-128E-Instruct`, which evaluates the full episode text and generates a scalar score from a standardized prompt. To ensure consistency, we fix the model temperature at zero. This method has been externally validated via a human-subject study on Prolific, which shows that episodes assigned higher scores by the model are also rated by participants as containing stronger cliffhangers.

System Instructions: You are a skilled literary analyst tasked with evaluating the strength of the cliffhanger in an episode of a serial novel.

Query: The strength of the cliffhanger is evaluated based on the following criteria:

- Tension and Suspense: How much does the scene make me anxious to find out what happens next?
- Emotional Investment: How much do I care about the characters affected by the cliffhanger?
- Surprise and Novelty: How unexpected or original is the cliffhanger?
- Stakes and Consequences: How significant are the potential outcomes for the characters or story?

Here is the episode text: {`EPISODE TEXT`}

For each criterion, provide a score from '0 = very low' to '10 = very high', 0 being the weakest and 10 being the strongest. Use the entire range of the scale and be mindful of differences between episodes. Then, calculate the average of these scores to get the final overall score.

Example Response: 6.5

We adopt three narrative shape measures introduced by Toubia et al. (2021) to capture structural variation in how stories are conveyed. *Narrative speed* quantifies how quickly the semantic content evolves within an episode—higher values indicate faster-moving narratives with quicker progression of ideas or events. *Narrative volume* measures the breadth of content covered, reflecting the number of distinct topics or themes introduced. *Narrative circuitousness* captures the extent to which the narrative loops back to previously discussed content rather than advancing linearly—lower values indicate direct progression, while higher values reflect a meandering flow. Figure W5.1 provides a visual abstraction of the three measures, taken from the original paper. The metrics are computed using OpenAI's `text-embedding-3-large` and publicly available code from `https://github.com/smoorjani/word_embedding_measures/`. All narrative features are standardized to have zero mean and unit variance.

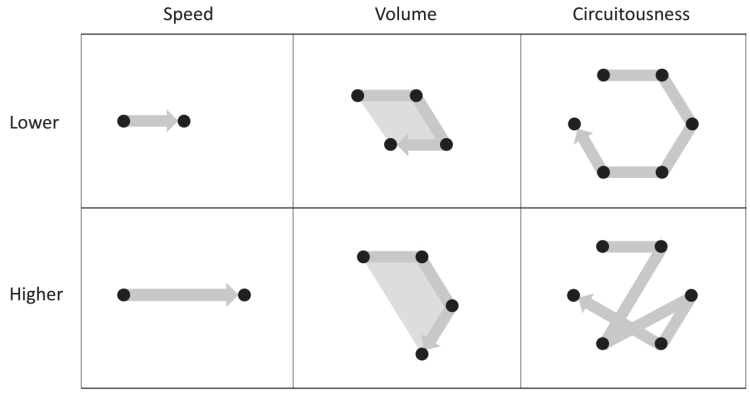

Figure W5.1: Narrative shapes from Toubia et al. (2021)

W6 Proximal Policy Optimization

As we follow standard PPO procedures, we keep the implementation details brief.

Advantage Estimation. To update the policy, we must evaluate how "good" each action is relative to the expected long-run value of the current state. This is captured by the *advantage* of an action:

$$A^{\pi}(s_t, a_t, z) = Q^{\pi}(s_t, a_t, z) - V^{\pi}(s_t, z),$$

which compares the expected return from taking action a_t at state s_t to the expected return from simply following the policy from s_t onward.

We estimate this quantity using Generalized Advantage Estimation (GAE; Schulman et al. 2015), which combines multi-period temporal-difference (TD) errors to reduce variance while maintaining accuracy:

$$\delta_t = r_t + \gamma V^{\pi}(s_{t+1}, z) - V^{\pi}(s_t, z) \tag{13}$$

$$A_t = \delta_t + (\gamma\lambda)\delta_{t+1} + (\gamma\lambda)^2\delta_{t+2} + \cdots \tag{14}$$

Here, r_t is the reward from Step S4, $V^{\pi}(s_t, z)$ is the estimated future value of being in state s_t under segment z, γ is the discount factor, and $\lambda \in [0, 1]$ controls the bias-variance trade-off.[13] The total empirical return used to update the value function, $V^{\pi}(s_t, z)$ is then $R_t = A_t + V^{\pi}(s_t, z)$.

Proximal Policy Optimization (PPO). The PPO algorithm updates the policy by minimizing a composite loss that balances three objectives:

$$\mathcal{L}_{\text{PPO}} = \mathcal{L}_{\text{policy}} + c_v\mathcal{L}_{\text{value}} + c_e\mathcal{L}_{\text{entropy}},$$

where c_v and c_e control the relative weight of each term.

[13] The value function V^{π} is a neural network embedded in the policy. We omit further discussion as it follows standard PPO implementation details and is not a novel aspect of our algorithm.

- **Policy loss:** Encourages the updated policy to increase the probability of actions with high estimated advantage, while preventing overly large changes:

$$\mathcal{L}_{\text{policy}} = \mathbb{E}_t\Big[\min\big(\Delta_t \cdot A_t,\ \text{clip}(\Delta_t, 1-\epsilon, 1+\epsilon)\cdot A_t\big)\Big], \quad \Delta_t = \frac{\pi_{\text{new}}(a_t \mid s_t, z)}{\pi_{\text{old}}(a_t \mid s_t, z)}$$

  The clipping threshold ϵ ensures conservative updates for training stability.

- **Value loss:** Ensures the value function accurately reflects the total expected return:

$$\mathcal{L}_{\text{value}} = \mathbb{E}_t\Big[\,(V^{\pi}(s_t, z) - R_t)^2\,\Big]$$

- **Entropy bonus:** Encourages the policy to maintain sufficient randomness in its decisions, aiding exploration and preventing premature convergence:

$$\mathcal{L}_{\text{entropy}} = \mathbb{E}_t\Big[\mathcal{H}\left(\pi(\cdot \mid s_t, z)\right)\Big]$$

W7 Hyperparameters and Training Schedule

The full training procedure is summarized in Algorithm 1, with model architecture details reported in Table W7.1. We train the model for 1,000 epochs, simulating 16,384 trajectories per epoch. The posterior network is held fixed after 500 epochs, by which point its segment assignments have stabilized. Batch sizes are set to 1,024 transitions for the discriminator, 128 trajectories and 512 users for the posterior network, and 128 transitions for the policy network. We use a discount factor of $\gamma = 0.99$ and optimize all networks using the Adam optimizer with a learning rate of 3×10^{-4} and default parameters ($\beta_1 = 0.9, \beta_2 = 0.999$). Policy optimization follows the standard PPO configuration from Stable-Baselines3 (Raffin et al., 2021). A full training run—covering all series and latent segments—typically completes in approximately five hours, depending on dataset size and rollout volume. All experiments are conducted with fixed random seeds.

Algorithm 1 IRL Algorithm for Serialized Content Consumption

Input: Initialize parameters for discriminator θ_0, ϕ_0, posterior ψ_0, policy η_0; observed trajectories $\mathcal{D}_{obs}$.
Output: Learned reward $f_{\theta,\phi}$ and policy π_η
for $i = 0, 1, 2, \ldots$ **do**

(S1) Trajectory Simulation: Simulate a batch of trajectories using current policy based on randomly sampled latent segment and series:

$$z \sim p(z), \quad \tau \sim \pi_{\eta_i}(\cdot \mid s, z)$$

(S2) Discriminator Update: Update $\theta_i, \phi_i \to \theta_{i+1}, \phi_{i+1}$ to minimize:

$$\mathcal{L}_{\text{disc}} = \mathbb{E}_{(s,a,s',z)\sim\mathcal{D}_{\text{obs}}}\left[\log D_{\theta_i,\phi_i}(s,a,s',z)\right] + \mathbb{E}_{(s,a,s',z)\sim\mathcal{D}_{\text{sim}}}\left[\log(1 - D_{\theta_i,\phi_i}(s,a,s',z))\right]$$

(S3) Posterior Update: Update $\psi_i \to \psi_{i+1}$ to minimize:

$$\mathcal{L}_{\text{posterior}} = \mathcal{L}_{\text{CE}} + \mathcal{L}_{\text{CR}}$$

(S4) Reward Assignment: Assign per-step reward for each transition in generated trajectories:

$$r_t = f_{\theta_{i+1},\phi_{i+1}}(s,a,s',z) - \log \pi_\eta(a \mid s, z) + \lambda_{MI} \cdot \overline{\log q}_{\psi_{i+1}}(z \mid \tau)$$

(S5) Policy Update: Update $\eta_i \to \eta_{i+1}$ using PPO with loss:

$$\mathcal{L}_{\text{PPO}} = \mathcal{L}_{policy} + c_v \mathcal{L}_{value} + c_e \mathcal{L}_{entropy}$$

end for

Model Component	Architecture Details
Discriminator Network	2-layer MLP (128, 64) with Leaky ReLU activations.
Posterior Network	2-layer bidirectional LSTM (hidden size 128) followed by 2-layer MLP (256, 64) with Leaky ReLU activations.
Policy Network	2-layer MLPs (128, 64) with Leaky ReLU activations.
Semi-supervised Autoencoder	Encoder and decoder: 2-layer MLP (512, 512) with ReLU activations; 32-dimensional bottleneck.

Table W7.1: Model Architecture

W8 Model Performance

We evaluate model performance by simulating consumption trajectories using the learned policy under both the training series and a hold-out set of unseen series. For each series, we simulate the same number of trajectories per latent segment as observed in the data and aggregate predicted outcomes to the series level.

In the training set, latent segments are inferred using the posterior network and held fixed during simulation, allowing us to assess how well the model replicates observed behavior within the estimation sample. In the test set, segment assignments are inferred solely from users' training-set trajectories and then applied to unseen series. This mirrors real-world deployment, where the platform must forecast behavior on new content without observing prior interactions. Test-set performance therefore provides a direct measure of the model's ability to generalize across narratives.

We focus on three model variants that isolate the key components of the framework: (i) `Our Model`, the full specification described in the paper; (ii) a `No Embeddings` variant that excludes unstructured content from the state space; and (iii) a `No Consistency Regularization` variant in which the posterior network is trained without the consistency loss described in Section 5.4.2. Performance is evaluated using four economically meaningful error metrics: aggregate mean absolute percentage error (MAPE) for total episodes read and total episodes purchased, and series-level mean absolute error (MAE) for purchase probability and completion rate.

Figure W8.1 summarizes the results. All three specifications achieve strong in-sample fit, indicating that structured variables combined with flexible heterogeneity are sufficient to replicate historical behavior within the training data. However, sharp differences emerge out of sample. The full specification maintains stable predictive accuracy on unseen series, with only modest increases in error relative to the training set. In contrast, the `No Embeddings` variant exhibits severe deterioration in test performance despite performing well in sample. For example, aggregate purchase MAPE increases from 6% in the training set to over 80% in the test set when content embeddings are excluded. This pattern indicates that, absent unstructured content, the model overfits to training-series regularities and fails to capture transferable, content-driven variation in demand.

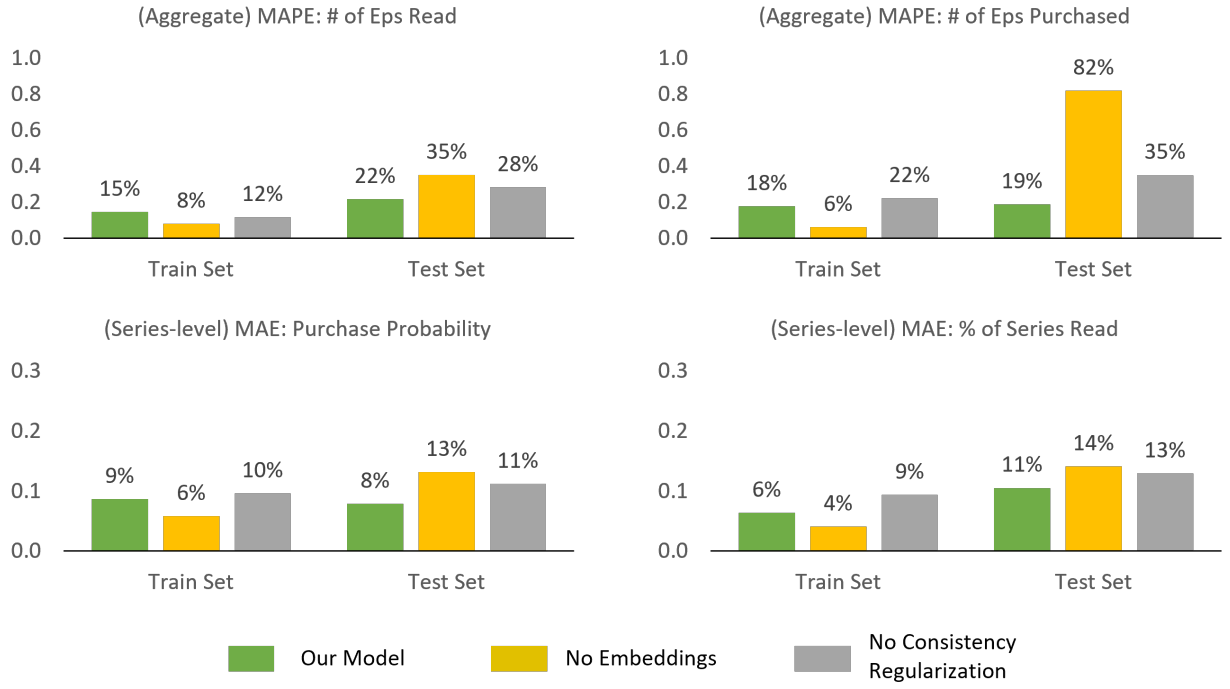

Figure W8.1: IRL algorithm performance for different model variants

The `No Consistency Regularization` variant also remains substantially worse than the full specification out of sample. Removing the consistency loss allows the posterior network to classify each trajectory independently, rather than learning stable user-level preferences. As a result, latent segments collapse toward coarse groupings based primarily on trajectory-level

outcomes—most notably the fraction of a series read—rather than economically meaningful trade-offs such as willingness to pay conditional on engagement.

This degeneration is illustrated in Figure W8.2, which reports aggregate consumption statistics by latent segment for the model trained without consistency regularization. Segments are largely ordered by completion rate, and purchase behavior varies weakly within segments. Consistent with this interpretation, 34% of users are assigned trajectories spanning all four latent segments under this specification, compared to only 11% under the full model. This instability weakens the model's ability to generalize behavior across series, even when content is observed.

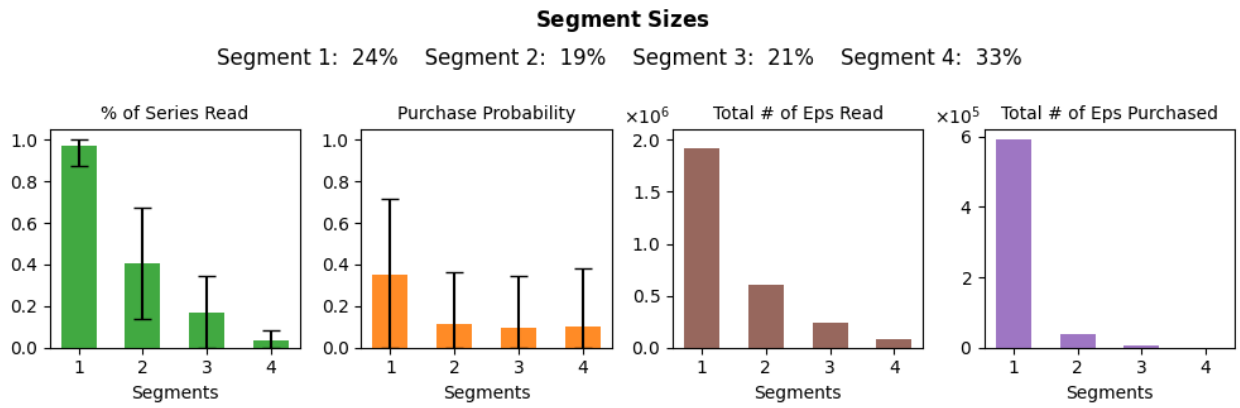

Figure W8.2: Aggregate consumption statistics across four latent segments when trained without consistency regularization. Latent segments are primarily distinguished by series completion rate rather than stable purchase behavior.

Overall, the performance comparison highlights the complementary roles of unstructured content and disciplined heterogeneity in modeling serialized consumption. Content embeddings are essential for capturing narrative-driven variation that generalizes across series, while consistency regularization ensures that latent segments represent stable user preferences rather than trajectory-specific outcomes. Together, these components enable the model to deliver accurate and robust predictions on unseen content.

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